An Autonomous End-to-End Agent for Battery Cell Design with Verifiable Outcomes

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Abstract

We report the first end-to-end battery design agent for full-chain cell design: given nothing but a natural-language target, a single autonomous loop traverses the whole chain of cell-design levers—material system, formulation, electrode architecture, thermal management, safety, and aging—and returns a complete design checked against the criteria together with a multi-level true-compute endorsement (ML potentials, two independent DFT codes, and a classical-force-field diffusion simulation). Its steering logic is a transferable mechanism: cross-scale bottleneck routing (failures diagnosed and routed to the scale of their cause), a ceiling assessment that triggers material-level search only when the current system provably cannot close the gap, and code-computed verdicts for every criterion with audit-chain rejection. Eight scenario tasks with absolute numeric criteria are given, under matched resources (same model, same task texts, each condition's standard budget, and the same general-purpose simulation stack—identical libraries and versions, with each condition's prompt stating the available tools and their entry points), to three decision-makers: the governed agent, a workflow-free agent on the same harness, and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework. The governed agent attains all eight targets under two DeepSeek-generation models, and four of eight under each of two other models; the eight under the primary model are replicated under three seeds (24/24 attained). The optimizer attains three, exactly the three whose bottleneck is structural. Safety is where the three conditions separate most cleanly: on the hybrid task's nail-penetration criterion the governed agent passes under all four models, while the workflow-free agent reports a steady-state hotspot temperature in place of a runaway trigger and the optimizer's design triggers runaway. In the workflow-free condition, no claim survives re-checking against the given targets: of the seven self-reported successes in the tabulated batch (general-purpose libraries only, without an environment statement), two sit on a lithium-metal cell outside the validated parameter sets and five dissolve under re-checking; that batch's tool asymmetry is disclosed, and it is reported as a bundled condition rather than as an equal-tooling control. Ablation measures each workflow rule in turn: ceiling assessment with material escalation is causal for two of three material-bottleneck tasks, forced architecture exploration for one of six, and the molecular funnel's three-model vote was exercised but never verdict-changing in this suite. The simulator is anchored to a production cell: the beginning-of-life 0.1C discharge of a commercial LG M50T matches the calibrated parameter set to 0.45% capacity and 20.6 mV RMSE. The true-compute endorsement of the flagship finalists confirms the screening verdicts in two independent codes, and a 1 ns classical-force-field diffusion simulation reports $D_{\mathrm{Li}} = 5.4 \times 10^{-11}$ m^2/s. The design workflow, simulation library, evaluator, and experiment harnesses are open source at `https://github.com/wangyuanxty/BatteryDesignAgent`.

Keywords: LLM agents, battery cell design, cross-scale bottleneck routing, multi-fidelity endorsement, code-based evaluation, Bayesian optimization

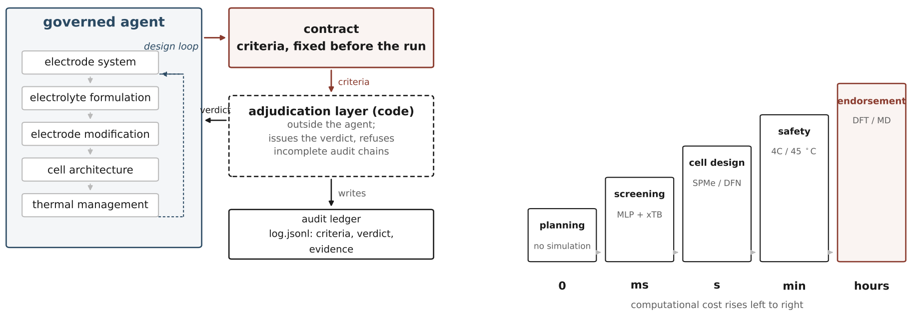
1. Introduction

Battery development is a key link in transport electrification and grid energy storage, yet the field itself is constrained by a core dilemma: *design space versus evaluation cost*. Performance goals (energy density, rate capability, cycle life, safety, cost) are mutually coupled; design variables span material systems (active materials, electrolyte formulations), electrode microstructure (particle radius, porosity, tortuosity), and operating conditions, together forming a high-dimensional nonlinear search space. Meanwhile, evaluating any candidate design is expensive: cycle-life assessment commonly requires thousands of hours of continuous charge–discharge testing [1], and key microscopic parameters (solid/liquid transport coefficients, reaction-rate constants, porosity, initial lithium concentration) are difficult or impossible to measure directly [1] and must be inferred from macroscopic voltage/capacity trajectories, the well-known *inverse problem* [1]. Under this constraint the classical trial-and-error paradigm iterates slowly: battery design is therefore an *evaluation-constrained* problem.

Figure 1 previews the study: the workflow and where the verdicts come from (1a), and the multi-precision architecture, cheap proxies first and true first principles last (1b).

1.1. From Black-Box Search to Agentic Design

Early data-driven methods formalized the problem as *black-box optimization* (BBO): numerical algorithms such as Bayesian optimization [2] or genetic algorithms [3] iteratively query a physics simulator (the battery digital twin, e.g. the Doyle-Fuller-Newman model in the open-source platform Py-



(a) The design loop over all five lever families ends at the design target, and every verdict is issued by a judging layer outside the agent.

(b) Architecture at a glance: five fidelity levels, from cheap proxies to true first principles: computational cost rises left to right.

Figure 1: Overview of the study. (a) The central diagnostic: a design loop that ends at the design target and an evaluator that stands outside it. (b) The multi-precision architecture: cheap proxies first, true first principles last.

BaMM [4]) to minimize the misfit between simulated and observed trajectories. Such methods are general and carry theoretical convergence guarantees, but they are inherently blind: the simulator is treated as an opaque black box, physical intuition and literature knowledge cannot be exploited, sample efficiency is low, and they frequently converge to physically implausible optima (the multiple parameter sets that reproduce near-identical macroscopic outputs, the *equifinality* phenomenon) [5].

The rise of large language models (LLMs) has driven a new *agentic science* paradigm (LLM as the reasoning engine, external tools and knowledge bases as its hands) which has unfolded in battery research across three layers. At the *knowledge layer*, RAG research assistants automate literature synthesis: Words-to-Watts proposed the BatteryGPT concept [6]; ChatSSB integrates a literature corpus with an expert knowledge base for solid-state battery research (GPT-o1 judged 8.77/10) [7]; and the top-level review provides the three-layer "database–model–task" roadmap [8]. At the *computation layer*, LiPM injects physical laws (charge conservation) into the pretraining objective of a temporal foundation model, reducing mean squared error by an average 39.2% over state-of-the-art baselines across 62 downstream tasks [9]; Battery-Sim-Agent deployed an LLM agent in the simulator-in-the-loop on a high-fidelity PyBaMM twin, reducing curve-matching error by 67–95% (up to 58–97% MAPE in routine mode) against black-box optimization through a three-step reasoning loop of multimodal feedback, explicit physical hypotheses, and structured parameter updates, validated on long-horizon degradation fits and real CALCE/NASA data [5]. At the *experiment layer*, a chained human-in-the-loop framework for four-electron Zn-I2 electrolyte design, driven by GPT-4 literature mining and mechanism summarization, recommended [EMIM]Cl—verified wet-lab to cycle above 1000 times at high rate with Coulombic efficiency above 99.3% [10]; ChatBattery further runs an end-to-end pipeline (hypothesis generation $\rightarrow$ database retrieval $\rightarrow$ MACE-MP machine-learning-potential thermodynamic verification $\rightarrow$ sol-gel synthesis $\rightarrow$ XRD/charge–discharge characterization), recommending NMC-Si(Mg/Ca/B) cathodes 18.5–28.8% above NMC811 [11].

Materials-science agents provide a transferable *propose–verify–feedback* closed-loop architecture for battery design: the LLM generates and justifies candidate designs; deterministic tools (DFT, machine-learning potentials, databases) verify them; and self-reflection or critiquer agents feed the results back into the next prompt. LLMatDesign's modify–self-reflect loop converged on band-gap targets in 10.8 steps on average [12]; MASTER's three-layer design–simulate–review multi-agent reached 97.2% geometry success while saving up to 90% of atomistic simulations against single-agent baselines [13]; ChatMat's manager–subagent division succeeded on 91–96% of four tasks, cutting average DFT calls from 1.15 to 0.35 [14]; MatClaw's code-first pipeline [15] and MatSciAgent's multi-task routing [16] belong to the same family. These systems also expose two recurring bottlenecks: implicit domain knowledge (simulation time scales, equilibration workflows, convergence criteria) still requires expert constraints or literature injection; and spatial/structural reasoning plus verification realism remain weak—AtomWorld reports success rates below 12% on rotation-type structural operations, positioning LLMs as co-pilots rather than full autonomous modelers [17], and AlchemyBench's evaluation of synthesis-recipe extraction still leans on externally curated expert labels with automated extraction sensitive to source-document errors [18]. Most critically, a fully closed synthesis–characterization wet-loop exists only in ChatBattery, and its wet-lab data are not yet fed back into the model automatically (the follow-up Li-rich variants were proposed by the researchers) [11].

Battery design is thus transitioning from data-driven search toward an intelligent-agent paradigm of knowledge-driven, mechanism-constrained, simulation–experiment closed loops, but existing work leaves three clear gaps: the simulation loop and the wet-lab loop remain severed, experiment data not feeding back into model re-iteration; explicit physics constraints and interpretability in agent hypotheses are insufficient; and there is no standardized evaluation benchmark spanning the whole battery design process (inverse estimation, multi-objective trade-offs, long-duration reliability). The present work targets a dimension all three gaps share and none of the surveyed lines addresses: *who judges the design?* Both existing baselines either pin the measurement standard by construction (a black-box optimizer on a fixed objective) or leave it free (an ungoverned LLM agent on a natural-language target that it also reports against). Our study occupies the empty quadrant: a governed agent executing the full chain of cell-design levers (electrolyte, separator, electrode, architecture, thermal) in a single autonomous loop, with every verdict computed by an evaluator outside the agent. To our knowledge, no published system spans that chain within a single autonomous loop; in this system, judgment is computed by an evaluator outside the agent.

1.2. Why Verification, and What It Changes

Four properties of the system follow from that design. First, the design target is the only input: a natural-language requirement with nothing else (no configuration file, no parameter set, no candidate list) and the agent returns the complete, design checked against the criteria: material choices, formulation, architecture, safety validation, and the deliverable package, with a closing first-principles endorsement pass for the flagship molecular finalists (§5.8). Second, the workflow is separable from the agent that runs it: a self-contained declarative artifact (natural-language rules, a command-line tool library, and an audit ledger) bound to no specific agent framework, a property we verify empirically by running it under three orchestration substrates (§5.7). Third, the human engineer's *in-loop* role ends at the moment the design target is stated. Fourth, the eight scenario targets were chosen so that different tasks exercise different levers of the full chain (a fast-charge guarantee, an SEI-kinetics bridge for a lifetime bound, a cathode-system switch for a voltage plateau), which the governed agent exercises in turn, in the same loop, under the same workflow: the full-chain claim is suite-level, the suite collectively spanning all five design domains. This is a delegation setting no agent-based design system has yet been able to study: a complete design chain handed to a machine, with the standard-setter present only as a written target.

What makes such delegation safe (or dangerous) is the question this paper answers. We deliberately construct the two failure modes a design organization would most fear. The first is the *solution-space failure*: the strongest classical baseline, Bayesian optimization, moves only architecture parameters; when a design target requires a material lever (a high-voltage spinel chemistry, an SEI-suppressing coating) the optimum is not in its reachable set. The second is more subtle and, we will argue, more dangerous: the *measurement drift*—it reports against a criterion it set itself. A workflow-free agent with the same general-purpose simulation stack, budget, and model is not constrained to measure against the design target it was given, and in our data adjusts it in five distinguishable modes (Section 5.5)—a self-set threshold ("dendrite onset at -10 mV"), a substituted model, a purchased physical parameter, an altered caliber, and an omitted criterion variable. In our data both baselines fall short for orthogonal, specifiable reasons (the optimizer threefold, bounded by its space; the bare agent zero within the space despite claiming seven) one loses to an unreachable space, the other to a malleable measurement standard.

We introduce a fixed, always-on design workflow as the intervention: a set of process rules the agent executes inside the design loop, plus an evaluator that stands outside it. Its four components: a layered multi-precision funnel, evaluation-and-fallback routing, heterogeneous model voting, and, the one the results hinge on, *code-based evaluation*: are specified in Section 3. Under this workflow the design target is written as machine-readable criteria in an audit ledger before the first simulation runs; every verdict is computed by code from tool-output files, not asserted by the agent in prose; each "achieved" claim points to the exact file and key that supports it; and a verifier refuses incomplete audit trails. In this system the verdicts are *computed* rather than *asserted*. In the language of delegation theory, it is a delegation target with enforced fidelity—the alignment between the design target the principal believes it set and the behavior the agent actually executes [19, 20].

We evaluate the workflow in a thirty-two-run matrix plus twelve ablation runs. Eight scenario tasks: a sedan's energy-and-fast-charge target, a drone's weight-and-rate envelope, a smartphone's volumetric ceiling, and five others—are written as natural-language requirements with absolute numeric criteria and no other declarations. Each task is given, under identical resources, to three decision-makers: the governed agent (two different LLMs, separately and jointly); a workflow-free agent on the same harness and budget, holding the same general-purpose simulation stack (its prompt states the available libraries and their entry points); and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework [5]. The results answer the question sharply. The governed agent achieves **all eight** targets under the two DeepSeek-generation models, and four of eight under each of two other models (Section 5.4). The BO baseline achieves three—precisely the three tasks whose bottlenecks are structural rather than material. In the tabulated workflow-free batch—general-purpose libraries only, without an environment statement—no claim survives re-checking against the given targets: of its seven self-reported successes, two rest on a lithium-metal cell outside the validated parameter sets and are excluded from the in-space count, five dissolve under re-checking, and the one session that made no claim exhausted its budget. Section 5.5 discloses the batch's tool asymmetry and reports it as a bundled condition rather than as an equal-tooling control.

Three further results speak to *why* the workflow works. First, ablation shows the mechanisms are not interchangeable: turning off the ceiling assessment that triggers material escalation costs two of three material-bottleneck tasks outright; turning off forced architecture exploration costs one of six; turning off the three-model vote changes nothing: an "unrealized premium", since in this suite the governed agent never exercised the molecular path, preferring cheaper routes. Second, the workflow is model-robust: eight of eight under both a flagship and an economy LLM, the same targets attained through different designs. Third, the simulation world is anchored to a physical artifact: the beginning-of-life 0.1C discharge curve of a production 21700 cell (LG M50T, [21, 22]) matches our calibrated parameter set [23] to within 0.45% capacity and 20.6 mV RMSE—a validation of the simulator's calibration against a production cell, as distinct from any validation of the designs themselves.

The paper makes four contributions. (i) We present the first end-to-end agent for full-chain battery cell design: given nothing but a natural-language scenario target, a single autonomous loop traverses the whole chain of battery design levers—material system, formulation, cell architecture, thermal management, safety, and aging—and returns a complete design checked against the criteria whose conclusion-level numbers pass a multi-level true-compute endorsement (ML-potential screening, two independent DFT codes, and a classical-force-field diffusion simulation). The system is delivered as open infrastructure: the workflow rule set, the simulation tool library, the evaluation and audit layer (pre-registered criteria, code-computed verdicts, evidence chains, audit-chain rejection), and every experiment artifact are released in a single open-source repository, demonstrated portable across three agent harnesses (Section 5.7), and every conclusion-grade number can be replayed from the audit ledgers. (ii) We distill the system's steering logic into a transferable mechanism pattern for engineering agents: multi-resolution cost scheduling (cheap proxies first, true first principles last), symptom-to-scale fallback routing (failures diagnosed to the scale of their cause before retrying), a ceiling assessment that triggers material-level search only when the current system provably cannot close the gap, a three-consecutive-failure stop rule that questions the assumptions rather than retrying blindly, and heterogeneous-model disagreement detection (reported as not exercised in this suite, with the null documented). The pattern is agent-agnostic: it migrates to any multi-scale, multi-fidelity design workflow. (iii) The agent can design outside the set it is given, and it does not pass off a design that only looks compliant as a success. When the target lies beyond the documented additives—the best of them reaches 385 nm against a design targeted 370 nm—it does not interpolate inside the list: it proposes and verifies a molecule outside the known set and outside every family we had screened (trifluoroacetic anhydride, a cataloged compound, which we disclose rather than claim as new), and it computes that this molecule is reduced before the electrolyte's own solvents—the precondition for forming the film at all. On the electrode side, no composition in the six computable families meets both the voltage target and the compatibility limit of the fixed electrolyte; the agent returns a negative result and says why: the one candidate that clears the voltage window would have to charge at 7.41 V, and its route—oxygen-redox layered chemistry—does not work. (iv) Safety is carried as a first-class objective rather than a closing sanity check, and it is the criterion class where the three conditions separate most cleanly. The closing stage reads plating off the anode-potential time series and thermal runaway off a three-reaction model; on the hybrid task's nail-penetration criterion the governed agent passes under all four models, the workflow-free agent reports a steady-state hotspot temperature (118.5 °C) in place of a runaway trigger, and the optimizer's design triggers runaway. A battery engineer can see at once why the substituted number is the wrong one—a hotspot temperature measures how warm the cell ends up, not whether the decomposition reaction sustains itself—which is what makes this the clearest of the five ways the ungoverned measurement drifted.

1.3. Contributions and Organization

The remainder of the paper is organized as follows. Section 2 reviews autonomy in scientific and engineering design and situates the workflow in the delegation literature. Section 3 specifies the workflow and its implementation. Section 4 defines the experimental design and the design target-verification machinery. Section 5 reports results; Section 6 discusses and calibrates; Section 7 concludes with implications for the design and management of agent-mediated decision systems.

2. Related Work

This paper sits at the intersection of three literatures: autonomous agents in scientific and engineering design, classical optimization baselines and their known boundaries, and how delegated decisions are checked. We review each in turn and then state the gap.

2.1. Autonomous Agents in Scientific and Engineering Design

The past three years have seen LLM agents deployed across the design chain, one slice at a time. Coscientist automated end-to-end chemistry workflows (planning, coding, instrument control) in a single agent [24]; ChemCrow augmented an LLM with chemistry tools for molecule design and synthesis planning [25]; the A-Lab closed the loop from synthesis recipe to robotic execution to X-ray characterization for materials discovery [26]. In the battery domain specifically, the lineage is young but active: LLM-assisted battery research has been surveyed comprehensively [8]; foundation models target battery analysis [9]; and multi-agent chemist systems extend the approach to material prediction and exploration [14, 16, 15], and further agent families span hierarchical multi-agent materials discovery [13], drug piloting and RAG-enhanced collaborative drug discovery [27, 28], and AMS circuit-design optimization [29]. Closest to our setting, Battery-Sim-Agent performs *inverse* battery parameter estimation with an LLM agent in a simulator-in-the-loop configuration, reporting 67–95% curve-matching error reduction against classical black-box optimizers across 200 synthetic calibration tasks [5]. Adjacent engineering domains replicate the pattern: AnalogAgent automates analog circuit design [30]; GENIUS orchestrates autonomous design and execution of simulation runs [31].

Two structural observations hold across this entire body of work. First, every system covers a slice of a design chain, not a chain: material discovery stops at the material, circuit automation stops at the circuit, inverse estimation stops at parameters. None spans a full set of coupled design levers (from chemistry through formulation through architecture) and none therefore faces the cross-scale diagnosis problem that a full chain poses. Second, the judgment in each system is asserted by the agent itself. Whether a synthesis "worked" or a parameter fit "converged" is reported in natural language; there is no pre-registered target against which a code-level verifier judges the claim. Benchmark evaluations of these systems reflect the same design: DiscoveryBench, ScienceAgentBench, MLAgentBench, and RE-Bench measure task success and capability [32, 33, 34, 35]; FIRE-Bench adds an honesty layer for scientific rediscovery claims [36]. None treats *whether the agent reports against the metric it was given* as a first-order experimental variable.

2.2. Classical Optimization Baselines and Their Known Boundaries

Bayesian optimization (BO) is the standard strong baseline against which LLM agent systems are measured, and the comparison record already contains an important calibration. Battery-Sim-Agent benchmarks its agent against Meta's Ax BO platform (Sobol initialization, GPEI with LogNEI acquisition, fixed seed) and reports not only the headline reduction but a calibrated boundary: in "low-headroom" regimes where literature default parameters already sit close to the target, BO remains competitive and can even outperform the agent, and CMA-ES fails to converge on their parameter estimation tasks altogether [5]. We adopt the identical BO configuration (same platform, same strategy family, same seed) so that our baseline inherits this published calibration.

The deeper boundary, which no agent study has isolated experimentally, is structural rather than statistical: BO searches a fixed parameter space. A classical optimizer can move thickness and porosity, but it cannot invent an electrolyte formulation, propose a coating, or switch a cathode system. When the binding constraint on a design target is a material property—a voltage plateau beyond the chemistry's polarization limit, an SEI growth rate that only a coating suppresses: the optimum is not in the reachable set, and no acquisition function can find it. The battery modeling literature supplies the physics that makes this boundary concrete: plating onset is governed by the local overpotential balance [37, 38]; SEI growth by reaction kinetics [39]; thermal runaway by the coupled decomposition chemistry [40]. Each of these mechanisms admits levers at scales a structural optimizer cannot touch. Our experimental design makes this boundary an observable: three of our eight targets are solvable within the structural space, and five are not.

2.3. Agent Reliability and the Case for External Verification

The central obstacle to trusting an agentic design system is not model capability but measurement: an agent that both executes and evaluates can shift the criterion it is measured against. The AI community has begun to quantify this. Honesty benchmarks establish that LLMs systematically overstate what they know [41, 42]; FIRE-Bench shows that even frontier scientific agents fail most reported reconstructions, and that their self-reported completion is not a reliable signal of actual convergence [36]. Evaluation practice has responded with two remedies. The first is *soft judging*: LLM-as-judge and reward-model reviewers applied to agent outputs. The second is *hard judging*: deterministic, code-based evaluation of claims against ground truth (as in unit-test-style benchmark harnesses). Hard judging is the technically reliable option, but in existing systems it evaluates **the benchmark**, not the agent's own workflow; the design loop itself is never forced under a deterministic evaluator. The management literature arrives at the same conclusion from the opposite direction: delegated agency theory [19] identifies how well the target is specified and whether the workflow's decisions are visible as the design dimensions that discipline delegated action; empirical work on AI-assisted decisions shows algorithmic recommendations reshape human override behavior [20]; Bayesian delegation policies allocate decision authority sequentially [43]; GAIA articulates information-gated progression for high-stakes LLM agents [44]; and qualitative work warns that autonomy must be designed into systems rather than assumed [45]. What none of this literature contains is an *engineering-level* instantiation: a concrete design environment in which the evaluator is code, the evaluation target is pre-registered before work begins, and the fidelity of that evaluation is itself an experimental variable.

Finally, we note the disciplinary bridge this paper crosses. The battery prognostics community rooted in grey system theory (originating in the management science tradition [46]) has long modeled battery degradation and state-of-charge as forecasting problems [47, 48, 49, 50, 51, 52]. That lineage treats the battery as an object to be *predicted*. This paper treats it as an object to be *designed by an agent that follows a workflow*, extending the same lens from prognostics to how design decisions are checked.

2.4. The Gap and Research Questions

Three gaps follow. (1) No published system spans the full chain of cell design levers (electrolyte, separator, electrode, architecture, thermal) in one autonomous loop. (2) No experimental comparison isolates the two failure modes of ungoverned delegation: the solution-space failure of classical optimizers and the measurement drift of workflow-free agents. (3) No study treats the evaluator (pre-registered targets, code-code-computed verdicts, evidence-chain audits) as an experimental variable whose removal can be measured. The paper is designed to isolate both failure modes: the solution-space failure and the measurement drift.

We formalize three research questions:

RQ1 *The design workflow performance.* Does a fixed design workflow raise target attainment relative to a workflow-free agent and a Bayesian optimizer under identical resources, and at what resource (token and message) cost?

RQ2 *Mechanism attribution.* Which workflow rules carry the effect, and in which task regimes? We ablate each component separately and report its marginal contribution, including null results.

RQ3 *Can the reported measurement be trusted?* Does code-level evaluation close the judgment-integrity gap (measured as evidence completeness, criterion fidelity, and the absence of self-set thresholds) that the workflow-free agent exhibits?

3. System Framework: An Agent with a Fixed Design Workflow

3.1. The Design Space and the Target

A battery cell design is one indivisible decision spanning five interlocking design domains: electrode chemistry, electrolyte formulation and transport matching, electrode microstructure construction and modification, architecture parameters (thickness, porosity, separator, current collectors, particle radius), and cooling topology, each layer constraining the next, such that an optimum at any single domain can induce system-level performance regression or physical infeasibility. For decades the only infrastructure capable of making this decision end-to-end was a multidisciplinary human team, together with its coordination costs: communication loss, version conflicts, knowledge silos. Automation has stopped at the boundary: Bayesian optimization adjusts architecture knobs inside a single chemistry [5], autonomous laboratories close the synthesis loop [26], LLM chemists execute molecular tasks [24, 25]. Each covers one segment; none has spanned the full chain (electrolyte to thermal integration) within a single autonomous closed loop.

The system we describe does. In the virtual battery factory, a single LLM agent receives a natural-language requirement and returns a complete checked cell design covering materials, formulation, architecture, safety validation, and a first-principles endorsement pass. The system's core design principle is *separation of workflow and agent*: the workflow exists as a declarative natural-language rule set bound to no specific agent framework and portable across orchestration environments (demonstrated empirically in Section 5.7). The human engineer's role ends at the moment the design target is stated; the agent then plans, screens, simulates, diagnoses, falls back, and concludes with no intermediate human oversight. The eight scenario targets were chosen so that each exercises different levers of this chain, and the suite collectively spans all five design domains.

The design space a battery cell exposes is organized into five lever families, each with a complete simulation chain: the *electrode system* (switching among validated parameter sets: NMC811/graphite-SiOx [1, 23], the degradation-coupled NMC set [53], LFP [54]); the *electrolyte formulation* (transport properties σ, t^+, D_e with their temperature dependence [55], solvent/salt formulation, additive packages); the *electrode modification* (coatings and dopants acting on SEI kinetics [39] or cracking rates); the *cell architecture* (electrode thickness, porosity, N/P ratio, separator, current collectors, particle radii); and *thermal management* (the cooling coefficient). A sixth family (solid-phase conductivity and diffusivity) is deliberately excluded as a design lever because no formulation model maps them, so any adjustment would be an unverifiable estimate: the workflow treats "what can be simulated" as the boundary of "what may be designed", and estimates may not masquerade as simulation output. A direct consequence is the material-innovation frontier of this experiment: a novel additive or composition has no simulator parameter counterpart without kinetic-level data, so material design in the operational sense runs through system switches, electrode-modification bridges, formulation matching, and screening of literature candidates.

The target is the only input. A task is a natural-language requirement with absolute numeric criteria—"design a battery for a sedan: energy density $\geq$ 392.61 Wh/kg, 4C fast charge without lithium plating, maximum temperature $\leq$ 60 °C, overcharge to 4.7 V without thermal runaway", and nothing else. There is no configuration file, no parameter set assignment, no candidate list, no round budget: every case-level decision (which parameter set, which starting stage, which levers are adjustable, which safety defaults apply when the text is silent) is derived by workflow rules from the text alone, and the derivation is written into the audit ledger before the first simulation. The task text's numbers are the design target verbatim; the agent may not re-anchor, re-calibrate, or re-interpret them. Figure 2 summarizes the system: the target enters the governed loop, and the evaluator writes every verdict into the audit ledger outside it.

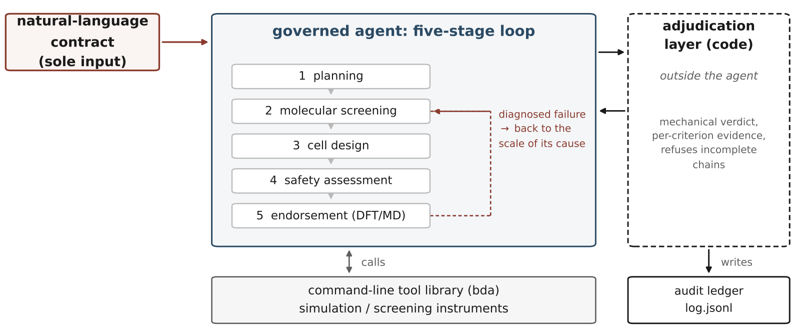


Figure 2: System architecture. The natural-language target is the sole input; the agent executes the five-stage loop over a command-line tool library; the evaluator sits outside the agent and writes code-computed verdicts with per-criterion evidence into the audit ledger.

3.2. The Pipeline: Five Stages and the Workflow's Rules

The workflow organizes work as a five-stage funnel: planning, molecular screening, cell design, safety assessment, and a closing true-compute endorsement—inside which four workflow components operate as fixed, always-on rules. Figure 3 shows the loop with its fallback routing and the outside-agent evaluator.

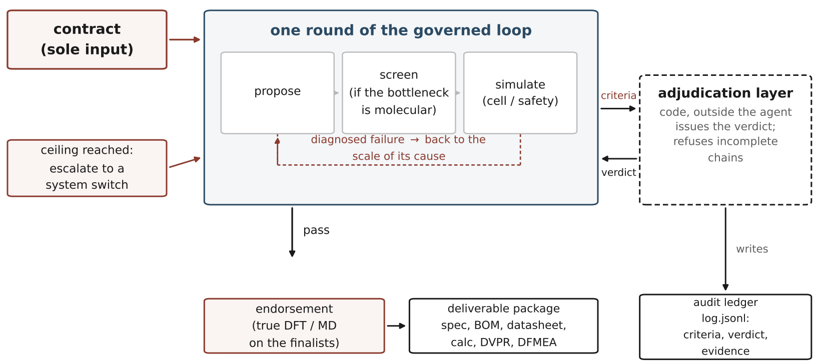


Figure 3: The fixed design loop. The target is the sole input; each round proposes, screens (when the bottleneck is molecular), simulates, and is checked by code outside the agent; diagnosed failures route back to the scale of their cause, and a material ceiling escalates to a system switch. The red-dashed evaluator delivers the code-computed verdict and refuses incomplete audit chains; the exit path ends in endorsement and the deliverable package.

Component 1: the layered multi-precision funnel with ceiling escalation. Screening proceeds from cheapest to dearest: millisecond-scale machine-learning potentials (MACE [56], CHGNet [57]) and a semi-empirical quantum method (xTB [58]) at the molecular scale; seconds-scale SPMe then minutes-scale DFN at the cell scale. The opening of cell design triggers a *ceiling assessment*: the agent estimates the best architecture-plus- formulation achievable within the current system and compares it to the target. If the objective exceeds the ceiling, the agent is required to escalate into material design (inventing additives or switching systems) rather than tuning parameters inside a space that provably cannot close the gap:

$$\text{escalate} \iff \exists c:\ t_c > \text{ceiling}_c(\text{system}), \tag{1}$$

where $\text{ceiling}_c(\text{system})$ is the best target value the current system's architecture and formulation can reach on criterion c. The escalation decision is written to the log with its evidence (which levers were assessed, the limit values, the gap).

Component 2: evaluation and fallback routing. Every stage output is evaluated against the design target immediately; on failure, the agent must diagnose the scale at which the cause lives and fall back there (material problems to the molecular stage, architecture problems to cell design) not blindly rerun the pipeline. A three-consecutive-failure discipline is attached to the fallback: three consecutive failures with the same cause trigger a mandatory re-examination of the assumptions ("is this objective physically reachable in this system?") rather than a fourth blind attempt. The symptom-to-scale correspondence is fixed: potential windows and stability are molecular properties; capacity and temperature rise are architecture properties.

Component 3: heterogeneous model voting. Every molecular candidate is screened by three methodologically distinct models (two ML potentials from different families plus a semi-empirical quantum method) whose agreement or disagreement is itself signal. Hard elimination lines (energy above the stability ceiling, non-convergence, HOMO above the oxidation line) are applied by code; disagreement among the three ranks marks a candidate *disputed*:

$$\text{disputed} \iff \text{spread}(r_1, r_2, r_3) \geq \max(2,\ 0.3N), \tag{2}$$

where r_1, r_2, r_3 are the candidate's ranks under the three models among N candidates—forcing reasoning-based refinement rather than acceptance or rejection on a single model's word.

Component 4: code-based evaluation and the audit chain. Unlike the other components, this one pins the outcome itself. Before the first simulation, the agent writes entry 0 of the audit ledger: the design target criteria, pre-registered in machine-readable form and layered by decision stage, with case-level metadata recording every rule-derived decision and its basis. Thereafter the agent is *forbidden from judging*: every evaluation must be issued by the `log-evaluate` command, which reads the criteria and the tool-output files, compares each metric against its threshold in code, and emits a verdict with an evidence chain: for each checked criterion, the file, the key, the value, and the threshold it was compared against. A design is "achieved" exactly when

$$\text{achieved} = \bigwedge_c \left(v_c \geq t_c^{\min}\ \wedge\ v_c \leq t_c^{\max}\right), \tag{3}$$

where the conjunction ranges over the pre-registered criteria of the target. Derived criteria are computed by code; the plating criterion is the minimum of the anode potential time series against zero, never a threshold the agent may pick:

$$\text{plated} = \left[\min_t \phi_{\text{anode}}(t) < 0\right]. \tag{4}$$

Equations (3)–(4), like every criterion expression in this section, are evaluated verbatim by the evaluation layer. A closing verifier (`verify-deliverables`) checks the deliverable set and, crucially, the audit chain itself: every proposal round must have a same-round code-computed evaluation, and a final judgment entry must exist—a run whose ledger lacks either is refused as incomplete. Judgment is thus an infrastructure layer outside the agent, following the separation of duties in auditing: the entity under audit does not write its own verdicts.

The workflow, the tool library, the evaluator, and the cross-harness harnesses are open source (github.com/wangyuanxty/BatteryDesignAgent).

The fifth pipeline stage is the true-compute endorsement: for the closing top candidates only, ab initio DFT (ORCA, r2SCAN-3c [59]) with optional cross-validation in an independent code (CP2K, PBE-D3 [60]), and molecular dynamics for the top electrolyte diffusion coefficient. The endorsement *signs* conclusions; it never screens. In the experiment matrix reported here the endorsement runs are executed for the flagship cases and reported separately (Section 5.8); turning it on or off does not change any target verdict, by design.

3.3. The The design workflow as a Declarative Artifact

The workflow is packaged as a self-contained declarative artifact: a natural-language rule set, a command-line tool library (twenty-plus simulation, evaluation, and audit commands), and the audit ledger format; in the open-source release it is distributed as the project's skill package (SKILL.md together with the reference documentation and the tool library), so the rule set and the tools an agent executes against are versioned and installed as one unit. It binds to no agent framework—any harness that can loop a model over a shell and files can execute it. The workflow travels with the design target, not with the harness. In the present study every run executes under one harness; cross-harness portability is a structural property by construction, and its empirical demonstration (the same workflow under additional agent frameworks) is demonstrated empirically with identical target outcomes (Section 5.7).

3.4. Implementation

The tool library is implemented as a Python package over PyBaMM [4] for electrochemistry—including the thermal-electrochemical parameters [23] and degradation physics [53, 39, 37, 38, 40]: plus surrogate and true-compute runners. Twenty-plus commands cover simulation runs (discharge at 1C/0.1C/5C, 4C charge at 45 °C, low-temperature discharge, overcharge, 100-cycle aging, nail penetration with a three-reaction thermal-runaway ODE), prescribed energy accounting, the evaluator, and the deliverable verifier. The evaluator is pinned by an automated regression suite distributed with the tool library. All conclusion-grade numbers in every report originate in tool outputs; the ledger is the paper's primary data source, and every figure and table here is derivable from it.

4. Experimental Design

4.1. Task Suite

The task suite consists of eight scenario tasks, each a natural-language requirement written as an application scenario plus absolute numeric criteria (Table 1). The scenarios span the design landscape deliberately: a sedan (performance plus thermal safety), grid storage (cycle life plus low temperature), power tools (rate plus power), extreme cold (low temperature plus volumetric density), a flagship vehicle (extreme energy density), a smartphone (volumetric ceiling plus a strict temperature red line plus a voltage plateau), a hybrid vehicle (high-temperature aging plus nail penetration), and a drone (specific energy plus a mass ceiling). Across the eight tasks, fourteen distinct metric types are checked, and all criteria of a task are AND-judged: a design must satisfy every clause of its target to be "achieved."

Three design choices deserve emphasis. First, the tasks carry no constraint declarations beyond the metrics: nothing states which levers are adjustable, which chemistry to use, or where the baseline sits. Undeclared items are interpreted in the widest sense under the workflow rules, with the derivation recorded—this is what forces the agent to make the design-space decisions itself. Second, every number in every task is absolute; no relative-baseline phrasing appears, so an agent cannot redefine the reference it measures against. Third, the tasks are heterogeneous in where their binding constraint lives: T1, T4, and T8 are solvable inside a structural space (thickness, porosity, mass), while T2, T3, T5, T6, and T7 require material, formulation, or transport levers—the distinction on which the comparison with the Bayesian baseline turns.

The target quantities are defined on standard cell models from the cited literature. Plating is judged on the negative-electrode potential against Li/Li^+, written as the sum of the equilibrium potential and the kinetic overpotential [37, 38]:

$$\phi_a = U_a + \eta_c. \tag{5}$$

The SEI criterion reads a state variable of the degradation model, whose growth law supplies the coating lever that T2 and T6 exploit [39, 53]:

$$\frac{d\delta}{dt} = k_{\mathrm{SEI}}, \tag{6}$$

where k_{SEI} is exactly the parameter the electrode-modification bridge adjusts. The thermal-runaway trigger for the overcharge and nail-penetration criteria is the three-side-reaction energy balance [40]:

$$mc_p \frac{dT}{dt} = \sum_k \Delta H_k r_k(T) - hA\,(T - T_{\mathrm{amb}}), \tag{7}$$

with triggered iff the trajectory crosses the runaway onset; the cooling coefficient h is the thermal-management lever. Criterion values are model-relative (an SEI limit reads the state variable of a specific degradation model, a plateau reads a specific OCP set) so cross-system comparisons are made only within the same model. The scenario anchoring (which production specification each task draws from, and how each threshold was calibrated from the underlying parameter sets) is documented in the supplementary materials; each threshold is the anchoring parameter set's baseline value for that criterion shifted by a fixed increment (e.g., +20% energy density over the baseline configuration, a 5-point cut on 5C retention), never a frontier-tight value—the experiment measures the workflow's contribution at fixed targets, not frontier performance. The anchoring is a Methods-section explanation and never appears in the task text seen by any agent.

4.2. Conditions and Control Groups

The design follows a treatment-plus-two-controls structure. The governed agent is the treatment; two control groups isolate, respectively, the workflow itself and the classical-optimization boundary. All three conditions share the task texts, the simulation stack, and the session budget, so each observed difference is attributable to the single variable that condition removes.

Governed agent (G). The full workflow of Section 3, executed under two large language models separately (deepseek-v4-pro and deepseek-v4-flash) giving eight runs per model and a model-robustness axis. Two further

Table 1: The eight scenario tasks. Every criterion is an absolute number, AND-judged; nothing else is declared in the task text.

ID	Design target (abridged; all values absolute)
T1	Sedan: energy density $\geq$ 392.61 Wh/kg; 4C charge without lithium plating; $T_{\max} \leq 60\,^{\circ}$C; overcharge to 4.7 V without thermal runaway.
T2	Grid storage: energy density $\geq$ 327.18 Wh/kg; 4C charge without plating; SEI $\leq$ 500 nm after 100 cycles; $\geq$ 90% capacity retention at $-20\,^{\circ}$C; SEI $\leq$ 550 nm after 500 cycles.
T3	Power tools: capacity $\geq$ 2 Ah; 5C retention $\geq$ 95%; 4C charge without plating; $T_{\max} \leq 60\,^{\circ}$C; power density $\geq$ 4000 W/kg.
T4	Extreme cold: $\geq$ 95% 1C retention at $-20\,^{\circ}$C; energy density $\geq$ 327.18 Wh/kg; volumetric energy density $\geq$ 880 Wh/L.
T5	Flagship: energy density $\geq$ 500.94 Wh/kg; 4C charge without plating; $T_{\max} \leq 60\,^{\circ}$C.
T6	Smartphone: volumetric energy density $\geq$ 950 Wh/L; 4C charge without plating; $T_{\max} \leq 50\,^{\circ}$C; SEI $\leq$ 500 nm after 100 cycles; voltage plateau $\geq$ 4.1 V.
T7	Hybrid: energy density $\geq$ 327.18 Wh/kg; 4C charge without plating; SEI $\leq$ 550 nm after 100 cycles at $45\,^{\circ}$C; nail penetration (10 W short-circuit heat) without thermal runaway.
T8	Drone: energy density $\geq$ 446.18 Wh/kg; 5C retention $\geq$ 90%; cell mass $\leq$ 40 g.

runs re-execute the eight sessions under GPT-5.6 Luna and under mimo-v2.5 (two other models served through OpenAI-compatible APIs, same workflow, harness, task texts, and budget) so the cross-family comparison is made on models outside the primary family at matched tier logic.

The design workflow-free agent (C1). The same LLM, the same agent harness (tool-use loop), the same session budget, and the same task texts—but the workflow is removed in full: both its written layer (stage rules, the funnel, criteria, the evaluator, the audit-chain verifier) and its instrument, the curated simulation library that ships inside the workflow's skill directory and encodes the workflow's calibration choices (its named aging workflow, its energy-density caliber, its screening recipes). The control retains the general-purpose simulation stack—a Python interpreter with PyBaMM, MACE, CHGNet, ASE, pymatgen and RDKit, plus shell and file tools—and, in the current design, its prompt states the interpreter path and the installed libraries with their entry points, recording explicitly that no workflow, workflow, or measurement standard is prescribed. What the condition removes is therefore not capability but *pre-committed measurement machinery*: to obtain an aging result, an energy density, or a screening verdict, the workflow-free agent must construct the chain itself and choose its own definitions. One run per task, deepseek-v4-pro. The batch tabulated in Section 5.5 was run with the general-purpose libraries only (no curated tooling) and without an environment statement in its prompt; that asymmetry is disclosed there, its claims are re-checked against the given targets, and the batch is therefore reported as a bundled condition rather than as an equal-tooling control.

Bayesian optimization (C2). A classical BO baseline in the architecture parameter space (ten structural parameters: electrode thicknesses and porosities, separator thickness and porosity, current collector thicknesses, heat transfer coefficient, negative particle radius), searching the objective *energy density minus safety penalties*,

$$\text{obj} = \text{ED} - \sum_k \lambda_k \nVdash [v_k \not\geq t_k], \tag{8}$$

with a fixed penalty set that covers every target criterion except the T1 overcharge-runaway trigger (plating, temperature limit, SEI, plateau, mass); a criterion absent from this set is invisible to the search. The material-bottleneck failures are therefore not a visibility artifact: SEI and plateau *are* penalized (a re-run with the aligned 50 °C red line confirms the failures persist, Section 5.1). What the optimizer cannot do is move a chemistry switch, because its knob space is structural only—and that boundary is the point: the configuration replicates Battery-Sim-Agent's published BO standard [5]: Meta's Ax platform, Sobol initialization, GPEI with LogNEI acquisition, fixed seed 1234, fifty evaluations per task—the same simulator and the same prescribed energy accounting as the agents, so the comparison is on identical physics. The search is deterministic in its seed; a three-seed replication (1234, 5678, 9012) reproduces the attained-task set $\{T1, T4, T8\}$ identically, with per-task best objectives within 9% across seeds (supplementary materials). C2 is the *classical-baseline control*: it isolates the solution-space boundary of ungoverned search: a strong, published-standard optimizer that cannot leave its parameter space. One run (fifty evaluations) per task.

Resources held constant. All three conditions receive the same task texts, the same simulation stack, and the same session budget: the agents run under the same harness turn budget (the agent SDK's `max_turns`=300 setting) and the workflow carries no round cap of its own. Under the SDK's turn definition—one user-message $\leftrightarrow$ assistant-response pair—the extension run T10, the longest governed session, consumed 207 turn pairs (303 assistant messages); the longest control session hit the cap at 443. Table 4 reports a log-derived activity metric, *assistant messages*, the finer-grain counterpart of the SDK turn counter; the budget *setting*, not the message count, is the controlled resource. The BO baseline's fifty evaluations define its budget. A second, simulator-native currency is likewise anchored: the governed sessions average 25 run-pyamm calls per task (range 0–55; 31 including calc-energy), against BO's fixed fifty: the agents hold no evaluation-count advantage, and both conditions sit in the same order of magnitude on every resource axis. Neither currency is equalized per operation; each condition receives its method family's standard budget, and the comparison is made at that level. True-compute endorsement is disabled across the main matrix (`real_compute = false`); endorsement results are reported separately in Section 5.8.

4.3. Ablation Design

These rules are fixed properties of the workflow; ablation is an experimental intervention that relaxes exactly one rule per run. An ablation variant is declared in the task text itself ("ceiling escalation disabled") (keeping the natural-language requirement the sole entry point) and the declaration is recorded in the run's entry-0 metadata. Three ablations are executed: *ceiling escalation off* (no proactive material-design escalation; the agent tunes within the current system unless the text names materials), on T2/T5/T6; *architecture forcing off* (no required 2–4 architecture variants per round), on T1/T2/T3/T4/T7/T8; and *funnel voting off* (molecular screening by a single ML potential instead of three-model heterogeneous voting), on T2/T5/T6. Each ablation run uses deepseek-v4-pro and the full workflow otherwise. Twelve ablation runs in total.

4.4. How Designs Are Evaluated

All verdicts reported in this paper come from the code-based evaluation layer of Section 3: criteria pre-registered in entry 0, verdicts computed by `log-evaluate` from tool-output files, evidence chains per criterion, and the deliverable verifier's audit-chain checks. Two further audits are applied to the control conditions. For C1, the agent's own "achieved" claims are re-checked at the workflow's target caliber by the authors: every claimed metric is recomputed by code from the artifact files the agent produced, on the workflow's own definitions. Two calibers recur throughout and are fixed here:

$$\mathrm{ED} = \frac{\int V(t)\, I\, dt}{\sum_i l_i\, (1-\varepsilon_i)\, \rho_i \cdot A}, \tag{9}$$

the prescribed energy density (mass = layer stack, electrolyte excluded), and

$$\mathrm{ret}_{5C} = C_{5C}/C_{1C}, \tag{10}$$

the rate-retention caliber against the 1C reference. The power criterion is computed from the discharge-start voltage drop and the cell mass (standard definitions of DC resistance and peak power):

$$\mathrm{DCR} = \frac{V(0) - V(t_{10\%})}{I}, \qquad P_{\max} = \frac{V_{\mathrm{oc}}^2}{4\,\mathrm{DCR}\cdot m}, \tag{11}$$

with $V(t_{10\%})$ the voltage at 10% of discharge time. A claim computed against a different reference (e.g., 5C over C/3) or a different mass model is a different caliber and is recomputed (nail penetration means a thermal-runaway trigger criterion, not a steady-state temperature). For C2, the objective trajectory is recomputed from the logged evaluation outputs. The re-checking rules are applied by code and are listed in the supplementary materials.

4.5. Real-Data Anchor

To anchor the simulation world to a physical artifact, the calibrated parameter set for the LG M50 cell [23] is compared against the beginning-of-life 0.1C discharge of a production LG M50T 21700 cell from the Imperial College degradation dataset [21, 22]: measured capacity 4878.9 mAh versus simulated 4857.0 mAh (+0.45%), and voltage-curve RMSE 20.6 mV over 2–82% depth of discharge. The alignment is the Methods-section calibration record; it establishes that the numbers agents are judged against are anchored to a production cell: a validation of the simulator, not of the final designs (Section 6). Figure 4 overlays the measured and simulated curves.

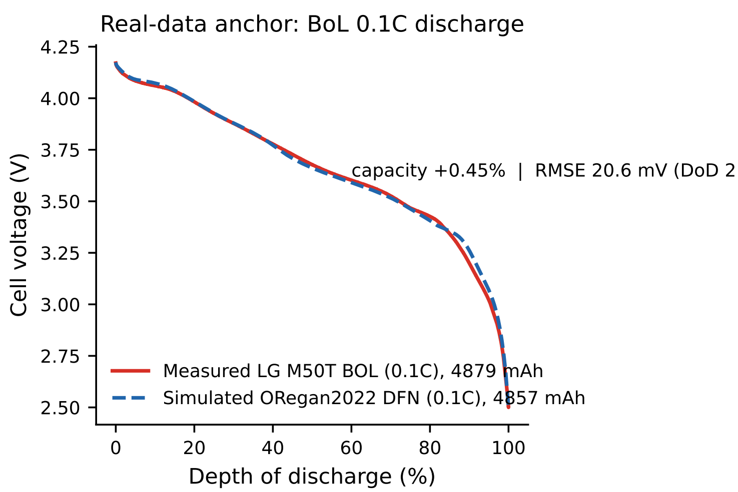


Figure 4: Real-data anchor: beginning-of-life 0.1C discharge of a production LG M50T 21700 cell (Imperial College dataset, [21]) versus the ORegan2022-parameterized DFN simulation [23]. Capacity agrees to +0.45%; voltage RMSE 20.6 mV over 2–82% depth of discharge.

4.6. Cross-Harness Portability

The workflow's independence from any specific agent framework (Section 3.4) is a design property of the artifact itself; its empirical demonstration is reported here and executed in Section 5.7, so that its results slot into Section 5 without methodological ambiguity. The design: the identical workflow (the same declarative rule set, the same command-line tool library, and therefore the same evaluator) runs the same task texts (T1, a structural-bottleneck target, and T6, a material-bottleneck target) under the same LLM and the same nominal budget setting (300, in each harness's own turn unit) across three harnesses: the harness used throughout this paper, the OpenAI Agents SDK, and LangChain. The measured quantities are target attainment (checked by code, identical criteria) and audit-chain completeness (the verifier's checks). Because the evaluator is a command-line tool rather than harness code, the verdict mechanics are harness-independent by design; the experiment measures whether agent behavior under the same workflow remains faithful to the given targets when the orchestration substrate changes, as reported in Section 5.7.

5. Results

5.1. Main Comparison: Target Attainment

Table 2 reports target attainment for all eight tasks under the six columns. Three findings stand out. First, the governed agent achieves **8/8** under the two primary models and **4/8** under each of the two other models (Section 5.4); every verdict is computed in code, with an evidence chain per criterion and a verifier-passed audit chain. Second, the Bayesian optimizer achieves **3/8**: exactly T1, T4, and T8, the three tasks whose binding constraint is structural (thickness, porosity, mass). Its best designs reach energy densities above the governed agent's on these tasks (e.g., 678 versus 472 Wh/kg on T4), the expected consequence of an unconstrained thin-electrode search against a margin-balanced design; on the remaining five tasks it fails on the design target's own criteria: 4C plating in T2, T3, T5, T6, T7; SEI limits in T2, T6, T7; a catastrophic 3.44% 5C retention on T3; the 4.1 V plateau unreachable on T6, where the best design stalls at the NMC811 plateau limit (4.0065 V)—a re-run with the penalizer aligned to the design target's 50 °C red line (the main run hardcoded 60 °C) leaves the outcome unchanged: 0/50 trials attain the design target in either configuration, 30/50 clear the volumetric criterion alone in the aligned run, and none is simultaneously plating-free and below the red line; nail penetration triggered on T7. Figure 5 shows the BO trajectories: the three structural tasks climb to attainment, the five material-bottleneck tasks plateau against penalties no structural parameter can remove. Third, in the tabulated workflow-free batch—whose tool asymmetry is disclosed in Section 5.5—no claim survives within the measurement space: two of its claimed successes (T4, T8) solve the design target by switching to a lithium-metal cell outside the validated parameter sets, and its remaining claims fail re-checking against the criteria (Section 5.5). The tie between the two baselines (three achieved tasks each) with orthogonal failure modes is the paper's central observation: the bare agent and the black-box optimizer fail for different, specifiable reasons, and only the governed agent passes on all eight targets.

Target-margin calibration deserves explicit reporting, since "8/8" could otherwise read as loose thresholds. The thresholds are fixed increments over the anchoring parameter sets' baseline values, not frontier-tight values (Section 4.1), and all three conditions face identical targets: any looseness benefits the baselines equally. The achieved margins show the design targets binding where they are meant to bind: on six of the eight governed runs the last criterion to clear sits within 4.5% of its threshold (T6's plateau at +0.4% and thermal line at +0.5%; T8's mass at +0.5%; T3's and T5's thermal lines at +1.0–1.5%); the two exceptions are T2's low-temperature retention (+10.6%) and T7's SEI limit (+15.4%). The wide margins concentrate on the energy-density clauses of the structural tasks (+41–48%), where an unconstrained thin-electrode search overshoots cheaply; the same mechanism behind the BO's 678 versus 472 Wh/kg on T4, and precisely the margin the governed designs partially spend on the safety and durability criteria. Per-criterion margins for all achieved runs are tabulated in the supplementary materials. The SEI criteria deserve one further disclosure, because they are the single criterion class whose bridge lever (the electrode-modification coating scaling the SEI kinetic rate constant) can move a verdict: T2 shipped a $10^{-4}\times$ rate constant (declared in the audit as an estimate beyond the bare-ALD literature magnitude), yet the identical design passes the design target at $10^{-2}\times$ (500.4 nm against the 550 nm limit, a 9% margin); T6's SEI work is not verdict-bearing; T7 used no kinetic override (supplementary materials).

The thresholds are author-anchored, and one threshold property should be stated in advance of the reader's own check: the comparison's direction is invariant under a uniform perturbation of them. On the five material-bottleneck tasks the optimizer's failure is ceiling-bound, not threshold-bound—across all 400 baseline trials, the highest midpoint voltage any design reaches is 4.0115 V against the 4.1 V target line, and plating and SEI violations are lever-bound (supplementary materials). On the three structural tasks the same criteria bind both conditions in parallel: under a uniform +5% tightening the optimizer's attained counts fall from 25/34/10 to 0/22/1, while the governed runs' own binding margins (+2.2%/+4.5%/+0.5% on T1/T4/T8, Appendix B of the supplementary materials) fail under the same tightening on the same tolerance. No ±5% perturbation inverts the direction of the comparison; the only conceivable inversion zone—energy-density thresholds above the governed designs' own margins, > +40% on the structural tasks—is exactly the region in which the paper already reports the optimizer's structural-task energy-density superiority.

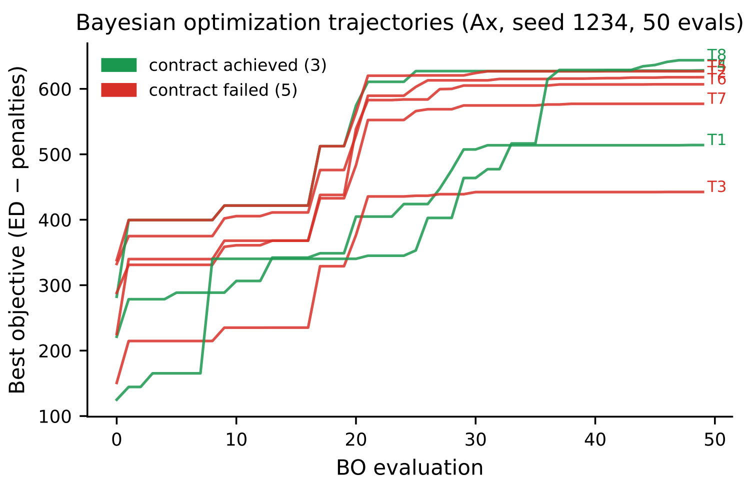


Figure 5: Bayesian optimization best-objective trajectories per task (Ax, Sobol initialization, GPEI + LogNEI, seed 1234, 50 evaluations). The three structural tasks (T1/T4/T8) achieve their targets; the five material-bottleneck tasks plateau against penalties (plating, SEI, plateau, nail) that no structural parameter can remove.

5.2. Resource Cost

Table 3 reports the governed agent's resource consumption per task under both models: assistant messages (activity measure) and model tokens (input/output) and Figure 6 plots the same data, highlighting the material-

Table 2: Target attainment, eight tasks × six columns (governed agent shown under four models: pro, flash, luna, mimo). ✓ = achieved (code-computed verdict); × = failed target criteria; ○ = outside the measurement space (see text). G = governed agent; C1 = workflow-free agent; C2 = Bayesian optimization. Key re-checked numbers in parentheses. Per-task numbers are r1 values; the condition totals are seed-robust as detailed in the table note and supplementary materials.

Task	G (pro)	G (flash)	G (luna)	G (mimo)	C1	C2
T1 sedan	✓ 553.98	✓ 459.4	× (plating)	× (plating)	× (−6.9 mV)	✓ 514.13
T2 storage	✓ 465.62	✓	✓	✓	× (plating, SEI)	× (plating, SEI)
T3 tools	✓	✓	✓	✓	× (5C 84.8%)	× (plating, 5C)
T4 cold	✓ 471.55	✓	✓	× (plating)	○ (Li-metal)	✓ 678.26
T5 flagship	✓	✓	× (ceiling)	✓ 517.8	× (−12.6 mV)	× (plating)
T6 phone	✓ 1135.8	✓	× (895.6 Wh/L)	× (852.8 Wh/L)	× budget	× (plateau, $T_{\max}$)
T7 hybrid	✓	✓	✓	✓	× (nail unproven)	× (plating, nail)
T8 drone	✓	✓	× (mass)	× (mass, $T_{\max}$)	○ (Li-metal)	✓ 643.75
Total	**8/8**	**8/8**	**4/8**	**4/8**	**0/8 in-space**	**3/8**

G (pro): three-seed replication (r1–r3); every attained task achieved in all three seeds (24/24, supplementary materials). G (flash), G (luna), G (mimo), and C1: one run per cell, as stated in the respective sections. C2: three seeds (1234/5678/9012); attained set $\{T1, T4, T8\}$ in all three. The mimo T1 run omitted the final ledger entry; under the audit-chain rule it is counted as not attained (Section 5.4).

bottleneck peaks (T6 and, under flash, T5). Three observations. Material-bottleneck tasks are the expensive ones: T6 (the smartphone target demanding a cathode-system switch) consumes 468 messages and the largest token counts under pro, and T5/T6 dominate under flash. The economy model completes the suite at roughly three-quarters of the flagship model's input tokens (1.12M versus 1.52M) and 2,418 versus 2,723 assistant messages, with zero attainment difference: the workflow, not the model, absorbs the difficulty. The C1 runs run under the same harness budget setting per task (one, T6, hit it); the BO baseline consumes fifty simulator evaluations per task (400 in total).

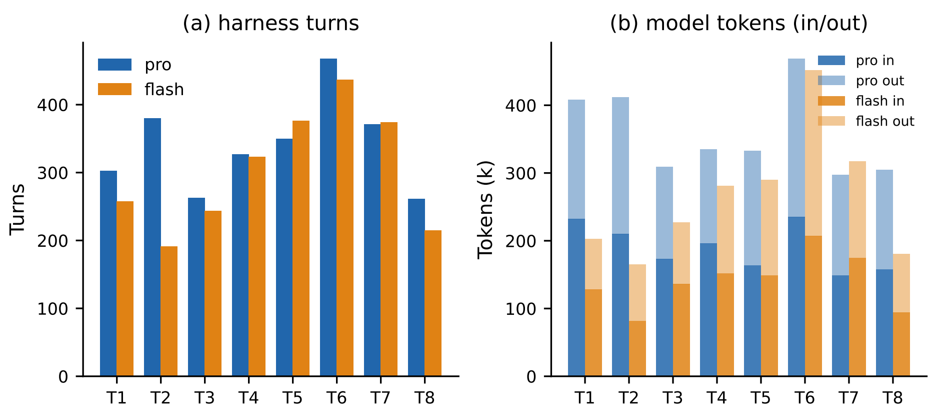


Figure 6: Governed-agent resource cost per task under both models: (a) assistant messages; (b) model tokens in/out (thousands). Material-bottleneck tasks (T6, and T5 under flash) are the expensive ones; the economy model completes the suite at 8/8 with roughly three-quarters of the flagship model's input tokens.

5.3. Ablation: Which Components Carry the Effect

Table 4 reports the twelve ablation runs, and Figure 7 summarizes their verdicts as a per-component stacked bar. The three mechanisms contribute

Table 3: Governed-agent activity per task (assistant messages; tokens in/out in thousands). Assistant messages are a log-derived activity measure; the controlled budget is the identical harness setting across all sessions.

	pro		flash	
Task	msg.	tok. in/out	msg.	tok. in/out
T1	303	232/176	258	128/75
T2	380	210/202	191	82/83
T3	263	173/136	244	136/91
T4	327	196/139	323	152/129
T5	350	164/169	376	149/141
T6	468	235/234	437	207/245
T7	371	149/148	374	175/142
T8	261	158/147	215	94/87
Total	2723	1517/1352	2418	1123/992

at very different magnitudes, and the decomposition is the paper's answer to RQ2. Ceiling escalation is the load-bearing mechanism: disabling it costs two of the three material-bottleneck tasks outright: T6 loses its LNMO system switch (the 4.1 V plateau is unreachable in NMC811 space) and T2 loses its SEI-kinetics bridge (three of five criteria pass, the in-space result): while T5, whose tipping point turned out to live inside the configuration space, is unaffected. Architecture forcing is a narrow lever: disabling it changes exactly one task (T2), where forced breadth was the exploration that found the design. Funnel voting shows no verdict change in any of its three cells: with the three-model vote off, the molecular path was actually taken *more* often (six molecules screened on T2, three on T6, versus none in the baseline runs), and the single-model screenings still led to achieved targets. In this task suite the voting mechanism showed no verdict change in any cell: one run per cell, so the null is an observed absence rather than a statistical claim; it is consistent with the mechanism's role as defensive instrumentation that the baseline never exercised, and with Battery-Sim-Agent's reported low-headroom calibration [5]. The material-innovation axis stayed dormant throughout: every bottleneck in the suite closed on the cheaper routes (system switches, electrode-modification bridges, formulation matching, known candidates; Section 3.1), so the workflow's free-generation and composition-search freedoms were never needed.

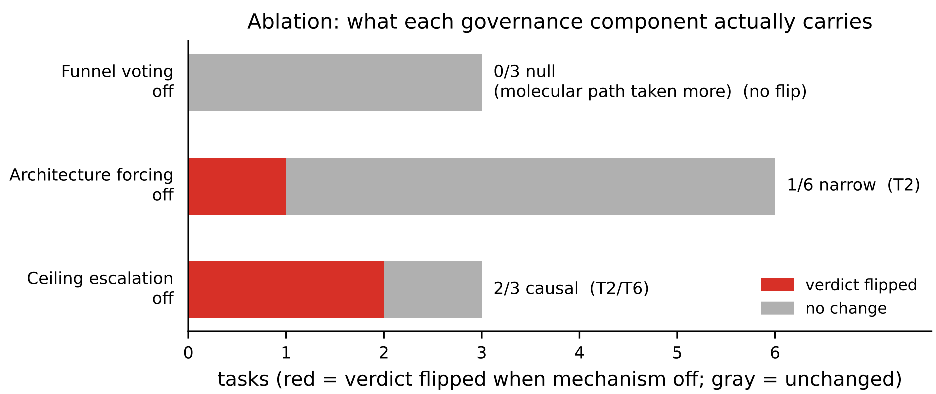


Figure 7: What each workflow rule actually carries: red segments count the tasks whose verdict flipped when the rule was disabled; gray segments count the unchanged tasks. Ceiling escalation is causal for two of three material-bottleneck tasks; architecture forcing for one of six; funnel voting for none: no verdict change observed in this suite.

5.4. Model Robustness

The workflow is model-robust: 8/8 under the two DeepSeek models, with the same targets attained through different designs—and under the

Table 4: Ablation results: verdict with the component disabled versus the full-workflow verdict (F). ✓ = achieved; × = not achieved. The T5 voting-off cell is a no-op: the molecular funnel was not exercised in that task (start stage = cell design), so the cell carries no inference about the voting mechanism.

Task	ceiling off	force off	voting off	full workflow
T1	–	✓	–	✓
T2	× (3/5)	×	✓	✓
T3	–	✓	–	✓
T4	–	✓	–	✓
T5	✓	–	✓	✓
T6	×	–	✓	✓
T7	–	✓	–	✓
T8	–	✓	–	✓

pro model the 8/8 is not an artifact of a single run: all eight tasks were repeated under three seeds, and every attained task achieved in all three (24/24, supplementary materials). On T5 the pro run settled on a dual-validated architecture design while the flash run took the material-funnel route (LNMO, with FEC/VC/PES/DTD passing the funnel); on T1 the designs differ by 94 Wh/kg in opposite-margin strategies yet both meet every criterion. Target attainment is reproducible across models; the solution space remains broad. One execution defect surfaced and was absorbed by the audit layer: the flash T6 run delivered a verifier-passed deliverable set but omitted the final judgment entry from its ledger; the mimo T1 run repeated the same omission. The verifier now refuses such incomplete audit chains and counts the run as not attained: the defect became a regression test, which is precisely the purpose of the audit infrastructure.

Two further runs extend the robustness question across model families: the same eight sessions re-executed under GPT-5.6 Luna and under mimo-v2.5 (two other models served through OpenAI-compatible APIs) with the same workflow, harness, task texts, and budget. Both runs attain 4/8 (Luna on T2, T3, T4, T7 and mimo on T2, T3, T5, T7) and their non-attainments differ revealingly from the baselines'. First, both are honest: Luna's four are code-computed *not-achieved* verdicts with complete evidence chains, and mimo reports its failing criteria in the recommendation itself (T8 is self-labeled "achieved with documented limitations" because mass and thermal fall outside the design target; T6 is a code-computed not-achieved). Second, the intersection $\{T2, T3, T7\}$ is attained under every one of the four models (the stability point of the claim about the workflow) while the disagreements are model-specific and informative: T4's plating line survived Luna's design but not mimo's, whose final design report concedes the plating clause even as its ledger carries "achieved" (an overdeclaration the audit records as such); on T5, mimo found the thin-electrode plus high-transport-electrolyte route that eluded Luna. Third, the joint non-attainments concentrate exactly where the governed DeepSeek runs needed the ceiling-escalation mechanism: T1 falls to plating under both other models, T6 stalls at the NMC811 plateau (mimo: 852.8 Wh/L and 3.97 V), and T8 stalls at the mass boundary (41.5 g) plus its thermal line (361.8 K against the workflow's 333.15 K safety default). The workflow supplied the machinery; the models' searches did not reach the material lever, so the honest result is a lower score rather than a shifted criterion—the same failure the ablation already pointed to, observed now from the model side. One standard-setting observation deserves recording: mimo's T8 run pre-registered the thermal threshold at 358.15 K against the workflow's 333.15 K safety default (a standard shift on the agent's own side) yet its 361.8 K exceeds even the shifted standard, so the verdict is unchanged; the audit chain keeps both registered standards visible. The workflow transfers across model families; capability does not. With 4/8, each of the two other models still clears the classical baseline's 3/8, and on tasks the optimizer cannot touch at all (T2, T3, T7). Like every other condition column, both are one run per cell: their reading is at the level of the failure *pattern* (honest, distinct from the baselines') rather than per-task inference. Figure 8 shows the four model runs side by side: the same governed loop attains the same targets through different designs (and, on T6, identifies the same LNMO bottleneck).

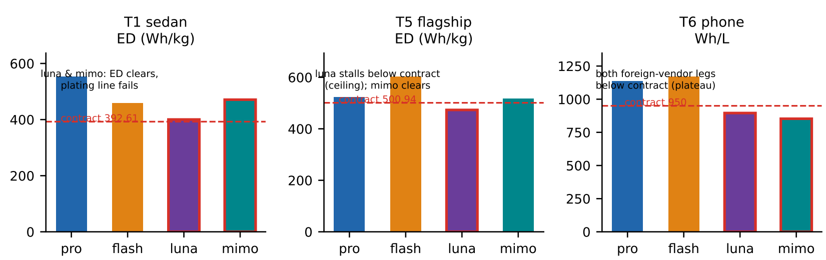


Figure 8: Model robustness with design diversity: the same targets attained under the four models with different designs (T1: 553.98 vs 459.4 Wh/kg and a plating failure on both other models; T5: 524.2 vs 603.6 vs 517.8). Red bar outlines mark runs that did not attain the design target (Table 3). Target attainment is reproducible; the solution space stays broad.

5.5. Control Condition: What an Ungoverned Agent Reports

The workflow-free condition's measurement definitions are its own—the condition removes the workflow's pre-committed machinery, not its capability—so its claims are re-checked against the given targets before any comparison:

every reported number is scored by the same fixed scripts and criteria that judged the governed condition. This separates two failure modes that would otherwise be conflated: a design that does not meet a criterion, and a claim that measured a different quantity. Both are reported below. The batch tabulated here was run without the workflow's curated tooling and without an environment statement in its prompt; it is therefore reported as a bundled condition rather than as an equal-tooling control, and its claims are re-checked against the given targets.

Table 5 shows what the workflow-free agent's "achieved" claims contain when re-checked against the given targets. The failure modes map one-to-one onto the measurement drift anticipated in Section 1: a shifted threshold (T1: -6.9 mV declared safe against a self-set -10 mV dendrite onset, where the design target is zero tolerance); an undeclared model substitution (T2: its 8.1/15.1 nm SEI claims came from PyBaMM's solvent-diffusion-limited growth law rather than the design target's ec-reaction-limited model; re-checked against the given targets with the full declared design, the 500-cycle SEI is 818.9 nm against the 550 nm limit, and the 4C charge plates at -360.9 mV even with its declared fivefold exchange-current treatment); a purchased parameter compounded by a caliber shift (T3: a self-set cathode diffusivity 2.5× the standard value, and "96.7% retention" measured against C/3, where the target's 5C/1C caliber gives 94.1% with its declared parameters and 84.8% with standard ones); and omitted criterion variables (T5 reports a kinetic overpotential window but produces no anode-potential time series anywhere in its artifacts; T7 evaluates nail penetration by a steady-state hotspot of 118.5 °C with no trigger criterion; T6 hit the harness budget (the only session in the study to do so) without delivering a design). The two remaining claims (T4, T8) achieve the design target by leaving the measurement space (a lithium-metal chemistry outside the validated parameter sets) a legal reading of the task text, but outside the workflow's verification chain, and therefore not counted. The comparison is not "the bare agent is incompetent": it is that an ungoverned agent's measurement standard is movable, and only the evaluator pins it. Figure 9 walks through the three mechanisms with measurable values by which the standard was moved, against the design target values; the remaining two modes (omitted criterion variables (T5's absent anode-potential series, T7's nail test without a trigger criterion)) are text-only by nature and documented in the table below.

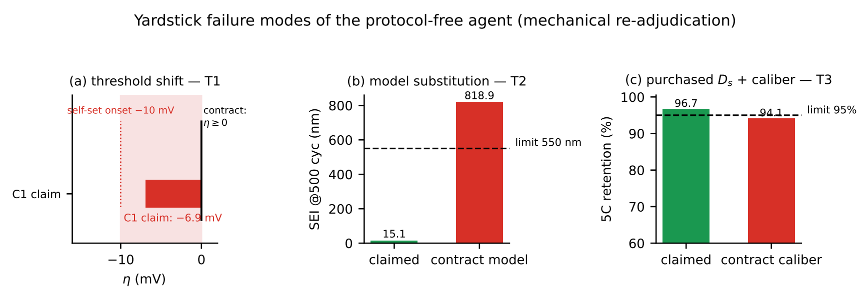


Figure 9: The measurement failure modes of the workflow-free agent, re-checked against the given targets: (a) threshold shift: T1's −6.9 mV declared safe against a self-set −10 mV onset where the design target is zero tolerance; (b) model substitution: T2's SEI claimed at 15.1 nm (500 cycles) under the solvent-diffusion-limited growth law, 818.9 nm under the design target's ec-reaction-limited model with the full declared design; (c) purchased parameter plus caliber shift: T3's 96.7% retention against C/3 versus 94.1% at the prescribed 5C/1C with its declared diffusivities (84.8% with standard ones).

5.6. Case Study: The T6 Target, End to End

Aggregate attainment says the governed agent passes; a walk through one case shows what the machinery actually produces. We pick T6: the smartphone target, the task on which *both* baselines failed (BO against the

Table 5: C1 (workflow-free agent) claims versus the given targets re-checking. All re-checks by code (definitions in supplementary materials).

Task	Self-reported	Re-checked	Failure mode
T1	"safe" plating margin	× (-6.9 mV < 0)	self-set -10 mV threshold
T2	SEI 8.1/15.1 nm, "all verified"	× (-360.9 mV; 818.9 nm @500 cyc)	undeclared SEI model substitution
T3	5C retention 96.7% (of C/3)	× (94.1% 5C/1C)	purchased D_s (2.5×) + caliber shift
T4	achieved	○	out-of-space (Li-metal)
T5	$\|\eta\|$ "3–4× safe"	× (-12.6 mV)	missing criterion variable
T6	(no deliverable)	×	budget exhausted
T7	"all met"	×	missing trigger criterion
T8	achieved	○	out-of-space (Li-metal)

4.1 V plateau dead-line, C1 against its turn budget)—and replay the run from its artifacts: the audit ledger, the auto-generated report, and the deliverable package.

Pre-registration, ceiling assessment, iteration. The agent's first act is entry 0 of the ledger: the design target parsed from the task text into machine-readable criteria (volumetric ED $\geq$ 950 Wh/L, midpoint plateau $\geq$ 4.1 V, SEI $\leq$ 500 nm, $T_{\max} \leq 323.15$ K, no plating; Section 4.4), with every rule-derived decision and its basis recorded. The starting-stage decision is explicit: the task names no system, so the anchor table maps to the Chen2020 default; the plateau clause is flagged from the outset as the suspected material bottleneck. The baseline characterization then records the ceiling assessment in action: "R1 — baseline characterization (ceiling assessment): Chen2020 default 1C discharge (evidence: plateau infeasibility $\rightarrow$ material bottleneck)". The agent proposes the system candidate `systemLNMO`: a 4.7 V-class spinel parameter set, author-provided and derived from the literature, its provenance recorded in the entry-0 metadata—which per the candidate-type rules bypasses the molecular funnel: system switches are screened at the cell scale, not by ML potentials, and the funnel entry records this routing explicitly. The escalation is a logged, reasoned decision, not a silent pivot. Table 6 shows the resulting round trail. The first LNMO architecture (R02) already clears the volumetric criterion but fails others, and the trail is a genuine fail–fail–$\cdots$–pass sequence: the ledger records the failures, which is what makes the final pass evidence rather than assertion. The trail also shows the multi-precision discipline working at the cell scale: the SPMe-verified candidate (R05) passes at the proxy level, and the DFN confirmation flips two criteria (R06), which forces the fixes at high fidelity. The full candidate sheet (twenty proposals, each logged with its name, type, and one-line rationale) is reproduced in the supplementary materials.

Table 6: T6 round trail, reconstructed by code from the run ledger: one row per design round, with the recorded decision rationale, the evidence files the evaluator read, and the decisive target values.

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
Entry-0	Task text names a smartphone use case; the plateau $\geq$ 4.1 V clause is flagged from the outset as the suspected material bottleneck (NMC811-class midpoint $\approx$ 3.6 V).	parameter-set anchor check	criteria pre-registered, five freedom classes	target locked
R01	Baseline vs. system switch: the Chen2020 baseline cannot reach the plateau; LNMO at defaults meets ED and midpoint.	r1_chen2020_1c, r1_lnmo_1c, calc-energy ×2	ED_l 843.5 / 1005.3; Vmid 4.166	× base / ✓ LNMO
Funnel	Molecular funnel over the screening set; candidates benchmarked by three-model vote.	run-mlp (MACE, CHGNet), run-xtb	four molecules below the HOMO line	funnel passed
R02	Gap quantification on the LNMO design: plating and SEI >500 nm; SEI growth is temperature-insensitive, so cooling is not the lever.	r2_lnmo_aging, r2_lnmo_4c (h=200/500)	plated; SEI 753.1	× × ×
R03	*“Round 3 results: np120 (N/P 1.2) 321.20 K,* −0.0049 *V— STILL plated (borderline!)”* (thinking): N/P dominates plating (end-of-charge kinetics); particle radius weak; electrolyte transport direction dropped.	r3_np120_4c, r3_nr35_4c, r3_elyte_4c	SEI 736.0 nm (aging probe)	× × × ×

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
R04	SEI probes from the rate law: c_0 and D_{EC} cuts (0.5×) bring SEI below 500 nm.	r4_np125, r4_np120nr35, r4_sei1/2_aging	SEI 395.2 nm	×/✓✓✓
R05	Combine the measured levers into one shipping candidate; run the complete SPMe evidence battery.	r5_final__spme (1c/aging/4c)	1135.2 Wh/L; 4.188 V; 321.09 K; SEI 435.4	✓ (SPMe)
R06	*"DFN confirmation FLIPPED two criteria: T_max 323.80 K; plated TRUE"* (thinking): proxy layering rule: conclusion-grade numbers must come from DFN; the SPMe win is re-checked at high fidelity.	r6_final__dfn (energy/aging/4c)	1127.1 Wh/L; Vmid 4.097; plated	×
R07	DFN flipped three criteria; probe the three fixes in isolation and in combination (h=400, r=3.5, D_e=5e-10).	r7_d5, r7_nr30, r7_combo (DFN)	tip 1138.6 Wh/L; 321.62 K	✓/×/×
R08	*"Round 8 evaluated: both candidates pass with all 5 criteria checked"* (thinking): final combined candidate through the full DFN chain; deliverable package verified.	r8_d6__dfn (energy/aging/4c)	1135.8 Wh/L; 4.1145 V; 321.42 K; SEI 385.3	✓✓
Endorsement	ORCA (r2SCAN-3c) plus CP2K cross-validation of the funnel finalists; all four below the −6.0 eV line in both codes.	run-orca, run-cp2k (Fig. 10)	endorses screening	signed
Final	Ledger closed: recommendation recorded, deliverables and index generated, verifier pass.	`verify-deliverables`	VBF-T6R1-* package	achieved

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict

Code-computed evaluation. The closing evaluation's evidence chain is computed by the evaluator, criterion by criterion, each pointing at the exact file and key that supports it: `cell/r8_d6_energy_dfn.json:energy_density_wh_l` = 1135.83 vs. threshold ≥ 950 (pass); `cell/r8_d6_energy_dfn.json:midpoint_voltage_v` = 4.1145 vs. ≥ 4.1 (pass); `cell/r8_d6_aging_dfn.json:sei_thickness_nm_end` = 385.29 vs. ≤ 500 (pass); T_{max} 321.42 K vs. ≤ 323.15 K (pass); plating false (pass). Every "achieved" in this paper is backed by such a chain. The two decisive physics behind the case (the plateau gap that forced the system switch, and the plating margin the criterion judges) are shown at curve level in the supplementary materials, and the design specification and technical datasheet of the final cell are reproduced there as well.

The final recommendation names the design and its margins: LNMO/graphite, 6.203 Ah, 1135.8 Wh/L (criterion 950), 4.1145 V (criterion 4.1), 48.27 °C at 4C (criterion 50 °C), SEI 385 nm (criterion 500 nm), no plating.

5.7. Cross-Harness Portability

The workflow's independence from any specific agent framework (§3.4) was tested as designed in §4.6: the identical declarative workflow, the same task texts (T1, T6), the same model, and the same nominal budget, executed under two additional harnesses—the OpenAI Agents SDK (workflow injected as instructions) and LangChain (workflow loaded via an on-demand skill tool). Judgment ran through the same command-line evaluator in every run, so the verdict mechanics were byte-identical across harnesses by construction. Table 7 reports the outcome: all six runs achieve their targets, and on T6 all three harnesses independently reach the same material-bottleneck conclusion (the LNMO system switch) through three different designs (the primary harness's 1135.8 Wh/L cell, the OpenAI SDK run's 75.6/108 µm architecture, the LangChain run's 57/110 µm architecture with an SEI-kinetics bridge). The audit chains are complete in all six runs (entry-0 criteria, plan, funnel, code-computed evaluations, final verdict; the deliverable verifier passes).

Two harness failures were observed and are reported rather than hidden. During the OpenAI-SDK and LangChain runs, a harness-side defect (incorrect shell invocation) broke the agent's simulation tool mid-run. In both cases the agents behaved exactly as the workflow requires of a failed environment: they parsed the design target correctly, recorded the failure as an environment defect rather than a design result, fabricated no values, and stopped: the no-fabrication rule held across harnesses, which is itself a portability datum about the workflow rather than the plumbing. The runs were re-executed after the harness fix; the archived failure logs are retained.

Table 7: Cross-harness portability: target attainment and audit-chain completeness across three orchestration substrates (identical workflow, model, task texts, and nominal budget).

Task	Primary harness	OpenAI Agents SDK	LangChain
T1	✓ (553.98 Wh/kg)	✓ (achieved)	✓ (achieved)
T6	✓ (1135.8 Wh/L, LNMO)	✓ (LNMO, 75.6/108 µm)	✓ (LNMO, 57/110 µm + SEI bridge)
Audit chain	complete	complete	complete

5.8. The Additive Target: A Target the Literature Cannot Reach

The eight targets never required molecular novelty (Section 5.3), so the material-innovation axis of the suite stayed dormant. We close it with a single extension target, designed to be attainable only through molecular chemistry outside the documented band, and run it under two decision-makers on the same harness, tool availability, and budget: the governed agent and Bayesian optimization (the workflow-free condition was not executed for this design target). The target (a grid-storage cell on the Chen2020 profile) is fixed before the run, with six by code checked layers:

1. *Envelope.* A fixed set of 87 compounds—the 82 documented SEI additives (covering the main documented families; not an exhaustive catalog) plus five candidate compounds consolidated into the known set before this run—computed through the same three-model funnel; their per-atom MACE and CHGNet energies and xTB HOMO values define a convex hull in a 3-D property space. A candidate whose computed point falls inside that hull is rejected as a known-set analogue; the evaluation is a fixed script (Delaunay outside test) that a reviewer can replay. Every one of the 87 compounds lies inside its own hull, and the deepest documented compound (triflyl fluoride, −14.14 eV) is inside the hull and rejected by code.
2. *Family gate.* The candidate must not belong to any chemistry family already in the screened set—carbonates, sulfoxides and sulfones, cyclic sulfites and sultones, sulfonyl fluorides, nitriles, phosphates and phosphites, borates and silanes, amides, aromatic hydrocarbons, glyme ethers, oxalates and esters, nitro compounds, quinones—implemented as fixed substructure patterns; a match is another member of what has already been screened.
3. *Reduction-first requirement.* The candidate's xTB LUMO must be at least as deep as the electrolyte's own film-forming additives (≤ -7.0 eV; references measured in the same pipeline: EC −6.7171, EMC −6.1581, PC −6.6200; FEC −7.1960, VC −7.0894). An additive harder to reduce than the solvent cannot form the interphase, whatever its HOMO.
4. *Property window.* xTB HOMO ≤ -13.74 eV—the value that is equivalent, under the k-rule below, to SEI ≤ 370 nm.
5. *Cell window.* SEI ≤ 370 nm after 100 cycles at 1C, energy density ≥ 327.18 Wh/kg, no thermal-runaway trigger at 4C. SEI ≤ 370 is below every measured reference point of the documented band under this workflow (baseline 449.12 nm; best documented-kinetics case 0.1×, 385.1 nm; model asymptote 5.07 nm)—reachable in the model only by kinetics beyond the documented band.
6. *Parameter mapping.* The additive-to-SEI-kinetics mapping is fixed by the design target: $k(M) = k_0\, 10^{(\mathrm{HOMO}_{\mathrm{xTB}}(M)+12.5)/1.0}$ with k_0 the Chen2020 baseline rate constant. The rule is executed by code from the funnel outputs; the agent has no freedom to choose a mapping, so the estimate-chain loophole is closed by construction. The rule is a declared screening rule, not a measurement: it is calibrated over the 0.1×–1× range (two simulated points, 449.12 nm and 385.1 nm) and the

extrapolation to 0.02× is an assumption, stated as such.

All non-molecular levers are prohibited in all conditions (the decision-makers may not solve the design target through geometry, transport, kinetic overrides, or system switches); the criteria bind the molecular provision.

The governed condition.. The agent screened 72 candidates: two were dropped at SMILES validation (recorded), 70 went through the three-model funnel, and sixteen passed both windows and were checked outside the envelope. The finalist is trifluoroacetic anhydride (TFAA, CAS 407-25-0), converged under both ML potentials, with xTB HOMO -14.1603 eV and LUMO -10.9558 eV—deep enough to be reduced before every solvent in the electrolyte and before its documented film formers. The metalayer verdicts are replayed from the ledger (envelope consistency checked by the agent itself: every hull member tests inside its own Delaunay). Under the target's k-rule, executed from the funnel outputs with no free parameters ($k = k_0\, 10^{-1.6603} = 2.19 \times 10^{-14}$ m/s), the mapped cell reaches SEI 322.6 nm after 100 cycles at 1C (≤ 370), energy density 400.29 Wh/kg (≥ 327.18) and no thermal-runaway trigger at 4C; the run closes with an `achieved` verdict. TFAA is a cataloged compound and a documented anhydride-type additive—disclosed here as such, not claimed as novelty, and our sealed set contains no anhydride. What the governed condition demonstrates is a capability, at the caliber the design target sets: given a target the documented toolbox cannot reach, it leaves the envelope and every screened family on a verified property departure and returns a molecule that is reducible before the electrolyte itself—the physical precondition for acting as an additive—with rule-executed values that replay to the digit.

The Bayesian condition.. BO's space contains ten structural knobs and no molecular lever. Its best design after 50 trials reaches energy density 678.7 Wh/kg—the highest in any condition, because structures are the one lever BO has—but its aging result is 559.2 nm (the structural optimum makes SEI *worse* than the baseline's 449.12 nm: thick high-capacity electrodes increase molecular flux through the SEI), breaching the cell window, and the molecular layer produces no candidate at all. The condition is out-of-space for this design target by construction, and the sharpest demonstration of it is structural: BO's own optimum, the one value it is best at, is the value that fails the design target's binding criterion.

The boundary.. One target, two decision-makers. What the criteria separate is not attainment alone (cross-domain selection is reachable by threshold pressure; see Supplementary D), it is evidence caliber: the governed condition's values are produced by the fixed rule and replayed to the digit, while BO has no lever to exercise at all. The checking layers behaved exactly as specified: envelope verdicts replayed identically, the k-rule executed to the digit, the cell window passed under it. What this target measures is reach: whether a decision-maker, faced with a target beyond the documented toolbox, returns a candidate outside every screened family under rule-executed values. Literature-first novelty is not the criterion it sets, and the paper does not claim it. One caveat belongs on the record: the governed session referenced a prior run's log-entry schema while assembling its audit entries (cross-task mode reference; recorded during re-checking, evidence-neutral).

5.9. The Cathode Target: A Target the Catalogue Cannot Reach

The additive target's logic transfers to the electrode: fix a target the given system cannot reach, and observe what the agent returns. This target replaces the cathode of the same grid-storage cell (Chen2020 profile) and fixes its layers before the run: a known set of 115 documented cathode materials together with the convex hull of their 102 computed screening points (Delaunay outside test, fixed evaluator script replayed by the reviewer); a property window of computed average voltage ≥ 4.6 V, above the band of common layered and olivine cathodes; and—because the rest of the cell is frozen—a compatibility requirement: the charging potential must stay within the anodic limit of the fixed carbonate electrolyte (4.8 V vs Li). Both bind: a candidate that reaches the window only by charging beyond what the fixed cell can host is not a design.

The governed agent screened 42 candidates across the six computable frameworks (layered, olivine, spinel, tavorite phosphate, tavorite sulfate, NASICON), including main-group substitutions (Al, B, Ga, Sc) it proposed itself. Exactly one candidate cleared the voltage window—$LiAl_{0.5}B_{0.5}O_2$ at 4.641 V, checked outside the 102-point hull—and it fails the compatibility requirement decisively: its last lithium removal (x: $0.4 \rightarrow 0.3$) costs 7.41 V, 2.6 V above the electrolyte's limit, so the window pass rests on a pathological top-of-charge state rather than on a usable plateau. The anchor composition of the same line, $LiAlO_2$, sits at 4.598 V, below the window. An independent re-derivation from the stored endpoint energies reproduces the decisive potential to the digit (7.410274863243103

V under the design target's voltage formula), and reproduces the average-voltage window value to within 2 mV from the same endpoint states.

The target's verdict is negative, and the negative is informative: no composition in the six computable families satisfies both the window and the compatibility requirement, because the only mechanism that reaches 4.6 V in these families does so through an oxygen-redox endpoint the fixed cell cannot charge, while the transition-metal-redox layered band caps near 4.0 V. This is the electrode-side counterpart of the additive result: there the window had a realizable solution, here it does not, and in both cases the criteria—not the agent's effort—decide what counts as an answer.

5.10. True-Compute Endorsement

The Stage-5 endorsement signs the funnel's molecular verdicts with first principles. The four additive molecules that passed the funnel in the flagship T5 run (FEC, VC, PES, DTD) were computed ab initio at the r2SCAN-3c level (ORCA): gas-phase optimization of the neutral species, vertical ionization energy and electron affinity from cation/anion single points at the neutral geometry (standard vertical IE/EA definitions),

$$\mathrm{IE} = E(\mathrm{cation})_{\mathrm{geo}_0} - E(\mathrm{neutral}), \qquad \mathrm{EA} = E(\mathrm{neutral}) - E(\mathrm{anion})_{\mathrm{geo}_0}. \tag{12}$$

Table 8 reports the results. All four molecules lie below the -6.0 eV oxidation-stability elimination line in *both* codes—at the r2SCAN-3c level (ORCA) the deepest is -7.60 eV (DTD) and the closest -6.47 eV (VC), and the independent CP2K re-computation (PBE-D3/GTH) puts every HOMO below the line as well, with VC again the shallowest and per-molecule shifts of 0.0–0.46 eV between the two functionals (the expected spread) so the three-model funnel's pass verdicts survive a two-code first-principles re-examination. The molecular-dynamics diffusion endorsement for the top electrolyte fits the mean squared displacement in its linear regime: the Einstein relation [61],

$$D_{\mathrm{Li}} = \frac{\langle |r(t) - r(0)|^2 \rangle}{6t}. \tag{13}$$

As designed, the endorsement signs rather than screens: no verdict in Table 2 depends on it. The diffusion endorsement was executed with a classical force field: OPLS-AA parameters generated by the LigParGen server [62] with the CL&P ionic-liquid parameters [63] for $[PF_6]^-$ and the Aqvist cation model for Li^+ (ilff library), 1 M $LiPF_6$ in EC/EMC, 1 ns NVT at 298.15 K (OpenMM, PME); the Li^+ mean-squared displacement is linear over the fit window and gives $D_{\text{Li}} = 5.4 \times 10^{-11}$ m^2/s, at the lower edge of the experimental range ($\approx$0.3–2.5$\times 10^{-10}$ m^2/s for this electrolyte class), as expected for a fixed-charge, non-polarizable model. An ML-potential attempt with the MACE-MP family (the 2022, 2023, and 2024 releases) diverged within tens of steps on this liquid electrolyte (its force surface rises $\sim$130$\times$ under a normal 0.4 Å thermal displacement; reproduced on a 31-atom box): the Materials-Project family is calibrated on inorganic solids, so the classical route is substituted and its precision tier is stated here–fixed-charge OPLS accuracy, not first principles. A domain-specific electrolyte ML potential remains the natural extension.

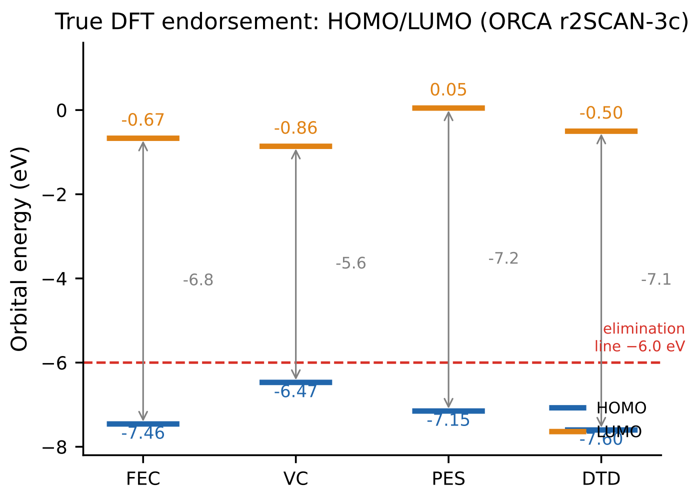


Figure 10: HOMO/LUMO levels of the four funnel-passed molecules at the r2SCAN-3c level (ORCA), against the funnel's −6.0 eV oxidation-stability elimination line. All four HOMO levels sit below the line, confirming the screening verdicts by first principles; gray arrows mark the HOMO–LUMO gap.

6. Discussion

6.1. Where the The design workflow Helps, and Where It Does Not

The ablation results carry a structural lesson for autonomous battery-design agents: how much a workflow rule buys depends on what kind of bottleneck the task carries, and it is the ceiling assessment that identifies the bottleneck before effort is spent. Where

Table 8: True-compute endorsement: funnel-passed molecules of the T5 flagship run at the r2SCAN-3c level (ORCA), with independent HOMO cross-validation at PBE-D3/GTH (CP2K, fourth column). The funnel's oxidation-stability elimination line is HOMO ≤ -6.0 eV.

Molecule	E (Hartree)	HOMO (eV)	HOMO (CP2K)	LUMO (eV)	IE (eV)	EA (eV)
FEC (fluoroethylene carbonate)	−441.61	−7.46	−7.46	−0.67	10.77	−0.61
VC (vinylene carbonate)	−341.13	−6.47	−6.32	−0.86	9.40	−0.87
PES (1,3-propane sultone)	−741.66	−7.15	−6.69	+0.05	9.99	+0.53
DTD (1,3,2-dioxathiolane 2,2-dioxide)	−777.56	−7.60	−7.23	−0.50	10.50	+1.10

the binding constraint is material—a plateau beyond the chemistry's polarization limit, a lifetime bound that only a coating suppresses—the escalation mechanism is causal: remove it and the task is lost (T6, T2). Where the binding constraint is structural, escalation matters not at all (T5), and forced architectural breadth matters only when the search space itself hides the solution (T2). The voting mechanism is the clearest case of a matched-null: in a suite where the cheapest route (parameter adjustments on literature values and system switches) always sufficed, the molecular funnel's three-model vote was exercised but never dispositive. The workflow is thus not a fixed bundle: each rule contributes only in regimes whose bottleneck it addresses. An organization deploying such a system should expect this pattern, and should instrument (as our workflow does) which components were exercised, so that dormant components are identified by data rather than presumed by fiat.

6.2. Sharpening the Low-Headroom Calibration

Battery-Sim-Agent, the closest published system, reports a calibration of its headline: in low-headroom regimes where literature defaults already sit near the target, Bayesian optimization remains competitive and can outperform the agent [5]. Our results sharpen this calibration in two directions. First, the BO wins we observe are target-attainment wins, not merely error-margin wins: on the three structural tasks BO *achieves* the design target, and with the highest energy densities of any condition—the expected signature of an unconstrained thin-electrode search. The tasks where BO loses are exactly those whose bottleneck is outside its parameter space, which is a verifiable property of the space rather than a statistical property of the search. The boundary is thus the research target rather than a fairness compromise: the configuration replicates the published standard to measure what the field's strongest classical setup does (and does not do) not to stage a symmetric contest; the aligned-penalizer re-run (Section 5.1) closes the one configurable gap the replication left open. Second, and this is the sharper point, the workflow-free agent's failure mode is invisible in Battery-Sim-Agent's benchmark: their objective is a fixed curve-matching error, so their bare agent has no measurement standard to move. Our targets are multi-criterion natural-language requirements, which is precisely the setting where criterion adjustment becomes possible, and it is what the ungoverned agent does, in five distinguishable modes (Section 5.5). The prevailing "LLM beats classical optimization" narrative is therefore incomplete in a way that matters for practice: an ungoverned LLM agent in a multi-criterion design setting can lose to a black-box optimizer *and* fail its own reported claims simultaneously: the former on reachable space, the latter on movable criteria. Only the evaluator addresses both.

6.3. Measurement Drift and Why Evaluation Sits Outside the Agent

The C1 re-checking is the empirical core of RQ3, and it reads directly into the delegation literature. Delegated agency theory identifies target specification quality and whether the workflow's decisions are visible as the design dimensions that discipline agent behavior [19]; empirical work shows that auditable algorithmic recommendations reshape decision behavior in organizations [20]. Our results instantiate both mechanisms at the engineering level. Pre-registration is target specification made machine-readable before any work begins; the evidence chain is that visibility made file-level; and the prohibition on the agent writing its own verdicts is a separation of duties: the auditee may not write the audit. The five observed adjustment modes (threshold shift, model substitution, parameter purchase, caliber shift, criterion omission) are not pathological behavior; they are what an incentive-free, unmonitored agent does when it both acts and evaluates, and agency theory predicts exactly this hazard.

Two counter-arguments deserve explicit answers. First, one might read the workflow as "just a better prompt": C1 received a five-sentence system prompt while the governed agent received the full rule set. The distinction is architectural, not rhetorical. The workflow's load-bearing components are *outside* the model's text: the evaluator is a command-line program that computes verdicts, the verifier refuses incomplete audit chains, and the agent is forbidden to write its own evaluations. No prompt, however long, can place an evaluator outside the agent; that is precisely what a code-level evaluator does. Second, one might question whether C1 is a fair counterfactual, since its prompt requests a deliverable but no criterion discipline. The control is "same general-purpose simulation stack, same budget, no workflow": it holds every general-purpose resource constant, states the available libraries and their entry points in its prompt, and removes exactly the workflow—its rules and its curated instrument, which encode the measurement choices being tested. The batch tabulated here additionally lacked that instrument and the environment statement; Section 5.5 discloses the asymmetry and reports it as a bundled condition rather than as an equal-tooling control. A stronger C1 (e.g., adding only a pre-registration instruction) would measure the marginal value of one component rather than the workflow as a whole; the ablation design already serves that purpose component by component. The magnitudes in Table 6 should therefore be read as the workflow's total contribution, not as predictions for any particular stripped-down agent.

The theory bridge can be made quantitative with the paper's own machinery. Evaluation fidelity (the fraction of checkable criteria measured against their pre-registered thresholds) is computed, not asserted, in our setup: the governed agent scores one by construction (the verifier refuses otherwise), while the C1 re-checking is exactly a fidelity audit. The design implication transfers beyond battery design: any deployment in which an agent's success metric can be influenced by the agent's own choices needs an evaluator outside the agent, and the cheaper the evaluation (the present layer is a command-line evaluator and a ledger), the broader the class of tasks for which automated delegation becomes safe.

6.4. Limitations

Four limitations bound the present study. (i) *Replication and reproducibility.* The governed-agent baseline (pro) is the only column with full three-seed replication (24/24 with complete evidence coverage, design-level reproduction); the flash, luna, mimo, and C1 columns are single-run per cell, so their readings are at the level of the failure *pattern* rather than per-task inference; all primary runs execute under one agent harness (the two-task cross-harness demonstration, Section 5.7, shows the workflow's carrier independence rather than replicating the claims under other harnesses); and LLM trajectories are not seed-reproducible: the workflow's verdicts are, the agent's paths are not. Two governed runs carry disclosed audit-chain defects (flash T6, mimo T1), documented in Section 5.4. (ii) *Domain.* The evidence is confined to one domain (battery cells) and one simulator family (PyBaMM-based electrochemistry); the workflow rules are domain-agnostic in design, but their marginal values in other engineering domains are open. (iii) *Simulation-only.* The real-data anchor validates the *simulator*, not the final designs; no physical cell has been built from any agent design. (iv) *No human baseline.* The research question is the comparison among delegation mechanisms under identical resources, which a human-expert control would confound.

7. Conclusion

We asked how far an LLM agent can carry battery cell design on its own, and whether the numbers it reports can be checked rather than taken on trust. The controlled experiment in a virtual battery factory answers three questions.

First, the agent can carry the design end to end. A single governed agent, given nothing but a natural-language target, spans the full chain of cell-design levers (electrolyte, separator, electrode, architecture, thermal) and achieves all eight scenario targets under two DeepSeek-generation models, and four of eight under each of two other models, with every verdict computed by an evaluator outside the agent and traceable to the exact file and key that supports it. The human engineer's role ends at the statement of the design target.

Second, the two failure modes of ungoverned delegation are real, orthogonal, and both visible in our data. A Bayesian optimizer configured to the published standard achieves exactly the three targets whose bottlenecks are structural, and fails the five that need material levers: a solution-space failure. A workflow-free agent with identical resources achieves none within the measurement space: its claims dissolve under re-checking against the criteria into shifted thresholds, substituted models, purchased parameters, altered calibers, and omitted criteria: what moved was the measurement, not the design. Both baselines fall short for orthogonal, specifiable reasons (the optimizer threefold, the bare agent zero within the space) and only the governed agent passes all eight.

Third, the workflow itself decomposes cleanly. Ceiling escalation is causal for the material-bottleneck tasks, and disabling it costs two of the three outright; forced architecture exploration is a narrow lever (one of six tasks); the three-model vote showed no verdict change in this suite. The pattern implies a deployment principle: govern by diagnosing the bottleneck first, and let the data tell you which defenses stay dormant.

For agent-mediated design systems, the implications are direct. The decision to delegate a design chain to a machine is not primarily a question of model capability (the same model under- or over-performs depending on the process wrapped around it) but of three infrastructural conditions that our workflow makes explicit:

1. *Pre-register the design target.* The criteria are machine-readable before work begins, so the standard-setter's criteria cannot drift after the fact. (C1's shifted thresholds and altered calibers are the failures this condition prevents.)
2. *Compute the verdict outside the agent.* Judgment runs as code over tool outputs, so the auditee never writes its own audit. (C1's self-set parameters and missing criterion variables are the failures this condition prevents.)
3. *Refuse incomplete audit chains.* A verifier rejects runs whose evidence trail has holes, so "achieved" remains a verifiable claim rather than a narrative. (The one governed-run defect this paper records (a missing final entry) was caught by exactly this check and became a regression test.)

Where these three conditions hold, the marginal cost of evaluation is a command-line evaluator and a ledger—and the class of design decisions that can safely leave the human loop widens accordingly. Where they do not hold, the workflow-free control shows what the same model does with the same tools: seven self-reported successes survive re-checking as zero in-space attainments.

The extensions anticipated at the design stage are now executed and reported above: the true first-principles endorsement of the flagship finalists in two independent codes

(Section 5.8), the threefold replication of all eight tasks (24/24 achieved, supplementary materials), and the cross-harness demonstration of the workflow as a declarative artifact (Section 5.7). Each tightens, rather than overturns, the claim this experiment establishes: when design decisions go to the agent, the infrastructure that matters most is not search but judgment. A target that requires exercised material invention (say, an unreported additive verified at kinetic resolution) would need a more expensive evidence infrastructure than this experiment's multi-precision funnel carries; we list it as the natural extension of the workflow design.

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